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By Francesco Filippucci, Peter Gal, Katharina Laengle and Matthias Schief

ABSTRACT/RÉSUMÉ

Macroeconomic productivity gains from Artificial Intelligence in G7 economies

The paper studies the expected macroeconomic productivity gains from Artificial Intelligence (AI) over a 10-year horizon in G7 economies. It builds on our previous work that introduced a micro-to-macro framework by combining existing estimates of micro-level performance gains with evidence on the exposure of activities to AI and likely future adoption rates. This paper refines and extends the estimates from the United States to other G7 economies, in particular by harmonising current adoption rate measures among firms and updating future adoption path estimates. Across the three scenarios considered, the estimated range for annual aggregate labour productivity growth due to AI range between 0.4-1.3 percentage points in countries with high AI exposure – due to stronger specialisation in highly AI-exposed knowledge intensive services such as finance and ICT services – and more widespread adoption (e.g. United States and United Kingdom). In contrast, the estimated range is 0.2 to 0.8 percentage points in countries where these determinants of AI gains are less favourable (e.g. Italy, Japan).

Keywords: Artificial Intelligence, Productivity, Technology adoption.

JEL Codes: C6, E1, O3, O4, O5.

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Évaluer les gains de productivité macroéconomiques de l'intelligence artificielle

L’article étudie les gains espérés de productivité macroéconomique liés à la diffusion des technologies d’Intelligence Artificielle (IA) sur un horizon de dix ans dans les économies du G7. Il s’appuie sur nos travaux antérieurs ayant introduit un cadre d’analyse allant du micro au macro, combinant des estimations existantes des gains de performance au niveau microéconomique avec des données sur l’exposition des activités à l’IA et les taux probables d’adoption future. Ce document affine et étend les estimations réalisées pour les États-Unis aux autres économies du G7, notamment en harmonisant les mesures actuelles des taux d’adoption des entreprises et en actualisant les trajectoires d’adoption future. Dans les trois scénarios envisagés, la croissance annuelle de la productivité du travail attribuable à l’IA est estimée entre 0,4 et 1,3 point de pourcentage dans les pays fortement exposés à l’IA – en raison d’une spécialisation marquée dans les services intensifs en connaissances fortement exposés à l’IA, tels que la finance et les services TIC – et où l’adoption est relativement plus répandue (par exemple les États-Unis et le Royaume-Uni). En revanche, cette fourchette est estimée entre 0,2 et 0,8 point de pourcentage dans les pays où ces déterminants des gains liés à l’IA sont moins favorables (par exemple l’Italie et le Japon).

Mots clés : Intelligence Artificielle, Productivité, Adoption des technologies.

Codes JEL : C6, E1, O3, O4, O5.

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Macroeconomic productivity gains from Artificial Intelligence in G7 economies

By Francesco Filippucci, Peter Gal, Katharina Laengle and Matthias Schief[1]

1. Introduction

Enhancing productivity growth is a key priority for G7 countries amid persistently weak productivity performance which weighs on per capita income and living standards. Artificial Intelligence (AI), with its rapidly advancing capabilities applicable in a range of contexts, is often seen as the next General Purpose Technology (GPT), on par with electricity or previous digital technologies such as computers or the internet (Filippucci et al., 2024). Although uncertainties exist regarding potential development paths of this technology ranging from a scenario of AI stagnation without major tech breakthroughs to an Artificial General Intelligence (AGI) revolution, the emergence of AI raises hopes for revitalised productivity growth (Samson, Zivkovic and Kalash, 2025). Historically, previous transformative GPTs often sparked periods of accelerated economic growth (Varian, 2019; Agrawal, Gans and Goldfarb, 2019; Lipsey, Carlaw and Bekar, 2005). To quantify the potential role that AI can play in reviving productivity in G7 countries, this paper builds on previous OECD work by Filippucci, Gal and Schief (2024) and presents estimates on the macroeconomic impact of AI on productivity over a 10-year horizon for G7 economies.

Predictions on the macroeconomic effect of AI on productivity differ widely across recent studies due to different modelling frameworks and assumptions feeding into the calculations, reflecting considerable uncertainty around them (Figure 1). For instance, Acemoglu (2024), and, building on him, Aghion and Bunel (2024), Bergeaud (2024) and most recently Misch et al. (2025), derive aggregate gains inspired by the task-based theoretical framework of Acemoglu and Restrepo (2018). These papers aggregate worker-level performance gains from AI in specific tasks – identified in the literature – to macroeconomic gains. In conjunction with relatively cautious assumptions regarding AI adoption and capabilities (the *exposure* of tasks to AI), Acemoglu (2024) predicts annual United States labour productivity growth driven by AI of around 0.1 percentage points (pp) over the next decade, while Aghion and Bunel (2024) find an effect of 1 pp, relying on more optimistic assumptions. Bergeaud (2024) and Misch et al (2025) predict about 0.4 and 0.3 pp respectively in their central estimates for the EU. Using a firm- and industry focused assessment, Briggs and Kodnani (2023) and McKinsey (2023) foresee 1.5 pp and up to 3.3 pp gains per year due to Generative AI, respectively. To put these numbers into perspective, during ICT boom in the

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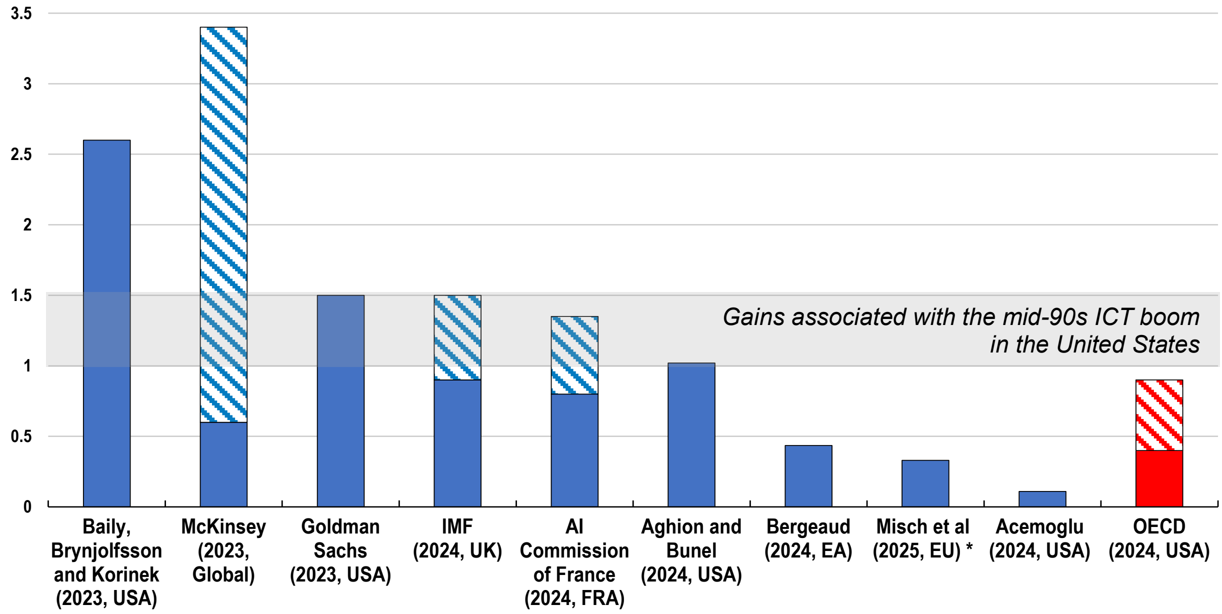
United States in the mid-90s the contribution of ICTs to annual labour productivity growth was estimated to be about 1-1.5 pp per year (Byrne et al., 2013; Bunel et al., 2024).

Figure 1. AI's predicted macro-level productivity gains vary substantially across studies

Predicted increase in annual labour productivity growth over a 10-year horizon due to AI, in percentage points*



Note: When the source presents a range of estimates as the main result, the lower and upper bounds are indicated by striped areas. In cases where modelling predictions primarily focus on TFP, labour productivity is obtained using simple assumptions about the aggregate capital multiplier (Acemoglu, 2024; Aghion and Bunel, 2024; Bergeaud, 2024; Misch et al, 2025). The estimates refer to the countries shown in brackets.
* Calculations in Misch et al (2025) refer to a 5-year horizon, presented in annualised form.
Source: See references at the end of the paper; for Goldman Sachs (2023), the underlying reference is Briggs and Kodnani (2023); for IMF (2024) the underlying reference is Rockall, Pizzinelli and Tavares (2024).

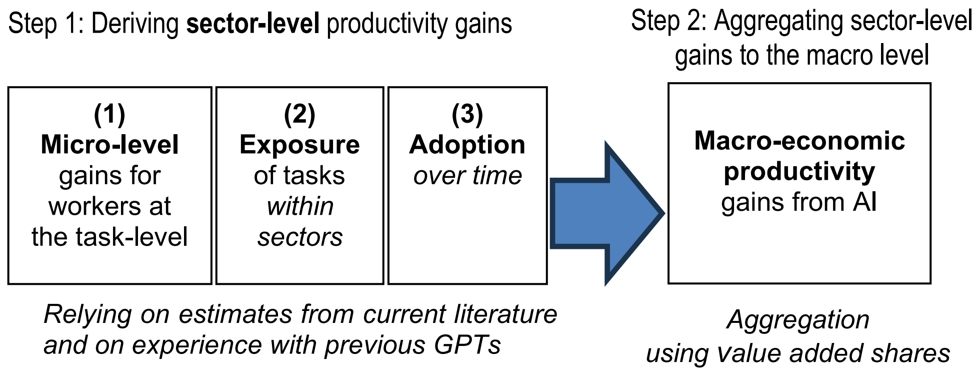
Recent OECD work (Filippucci, Gal and Schief, 2024) builds a micro-to-macro framework where sectors play a key role. It combines existing estimates of micro-level performance gains with evidence on the exposure of activities to AI and likely future adoption rates, applying Acemoglu's (2024) framework to sectors. In addition, it also relies on a multi-sector general equilibrium model with input-output linkages borrowed from Baqaee and Farhi (2019) to arrive at aggregate effects and predicts that AI's contribution to annual labour productivity growth in the United States ranges from 0.4 to 0.9 pp over a 10-year horizon.[2]

This paper refines and updates these previous calculations. It uses more recent and more detailed data on rapidly evolving AI adoption and aims to harmonise them across countries. In particular, the paper estimates macroeconomic productivity effects from AI in two broad steps (Figure 2). First, it builds on Acemoglu (2024) and considers a range of estimates from the growing literature on (1) **micro-level productivity gains** from AI at the task level, (2) the **exposure** of tasks within sectors to AI and (3) predictions of **future AI adoption** across firms in each sector under different scenarios. The second step

[2] The choice of a 10-year horizon is motivated by mainly two factors. First, the debate on the productivity effect of AI highlights a so-called productivity J-curve arising from an underestimation of productivity growth in the early years when a GPT is introduced. Second, this avoids considering more speculative scenarios for the very long term, including the possibility of faster innovation and explosive growth (*singularity;* Nordhaus, 2021; Aghion, Jones and Jones, 2019; Trammell and Korinek, 2023).

aggregates these sectoral gains using country-specific shares of different sectors in terms of their contribution to GDP (*value added* shares) to arrive at macroeconomic effects.[3]

Figure 2. The framework for aggregating AI's productivity gains from micro-to-macro



Source: Adapting Filippucci, Gal and Schief (2024).

C*urrent* and *future* AI adoption among firms and their cross-country differences are key determinants of AI driven productivity gains. However, challenges arise due to the lack of harmonised data, even across G7 economies, on high-intensity, regular use of AI in core business functions and operations – in the production of goods and services – which matter most for assessing productivity gains.[4] To address this data gap, this paper aims to harmonise *current* AI adoption across countries wherever possible and rely on regression-based predictions – driven by digital infrastructure and skills – for countries where comparable measures are not available (United Kingdom and Japan).[5]

A close examination of Eurostat's survey on AI use among firms reveals that AI adoption in core business functions by businesses is substantially lower than AI use for any business purpose: in 2024, the respective figures are about 3% and 14% in Europe. Private use among, individuals, which is often the focus of public commentaries when discussing the fast spread of latest Generative AI (e.g. large language models) is much higher still, reaching 39% in the United States according to a recent large-scale survey (Bick, Blandin and Deming, 2024). After harmonisation and further adjustment steps that aim to account for country-level

[3] Specifically, the second step aggregates sector-level gains to the macro-level by summing sectoral TFP gains, with each sector weighted by its current value-added share in aggregate GDP, which provides a first-order approximation to the aggregate TFP gains, as implied by Hulten's (1978) theorem. Filippucci, Gal and Schief (2024) go beyond this approximation using a multi-sector general equilibrium model, which allows to assess how changes in the sectoral composition of the economy can act as a drag on aggregate growth (Baumol's growth disease; Nordhaus, 2008). Given the relatively limited role of this latter effect in driving the overall productivity gains found in Filippucci, Gal and Schief (2024), Baumol-effects are discussed only in general terms in this paper (Section 3, Box 1).

[4] For instance, the question of the **United States Census Bureau** survey on AI use reads as follows: *"During the last two weeks, did this business use Artificial Intelligence (AI)1 in producing goods or services? (Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.)"* in conjunction with the answer options *"yes, no, and do not know"*, while the question of the **Eurostat** survey is *"Does your enterprise use Artificial Intelligence software or systems for any of the following purposes?"* in conjunction with the answer option *"[…] Use of AI for production or service processes",* among several other options. An important harmonisation step in this paper is to retain the responses only to this option, to focus on high intensity AI use in core business functions also in Europe, similarly to the United States and Canada. See more details in Section 2.3 and the Annex B, Table B.1.

[5] Consequently, estimates for the United Kingdom and Japan are surrounded by higher uncertainty than it is the case for other countries.

differences in a few areas of AI adoption determinants (digital infrastructure and skills), the resulting estimates of our preferred AI adoption rates among businesses ranges from about 2% to 6% in 2024 across the G7: they are highest in the United States, followed by Canada, the United Kingdom and Germany while it is lower in France, Italy and Japan.

These estimates of *current* AI adoption combined with the expected pace of the spread of the technology are a key indicator regarding *future* AI adoption. Central to the projection of future adoption rates is the adoption trajectory of previous GPTs. These are characterised by an S-shaped curve: an initially accelerating diffusion followed by a slowing down as the use of technology is becoming widespread, with the shape of the curve determined by various parameters. For the case of AI, these parameters are inspired by the continued increase in adoption speed across successive previous GPTs, such as electricity, followed by information and communication technologies (ICT) tools (including personal computers and the internet) and more recently mobile phones - capturing slow, medium, and fast adoption pathways. In conjunction with assumptions on different technological capabilities of AI – baseline vs. expanded capabilities[6] – the various AI adoption speed assumptions form a key component of the different scenarios to quantify AI's macroeconomic productivity effects.

Results on macroeconomic productivity gains indicate that AI's contribution to annual labour productivity growth are expected to vary significantly across G7 economies (as discussed in detail in Section 3, Table 2 and Figure 15) Under the most pessimistic scenario (low AI adoption, baseline AI capabilities), annual labour productivity gains range from about 0.2 percentage points in Japan and Italy to approximately 0.4 percentage points in the United Kingdom and the United States over the next decade. In the central scenario (medium adoption speed, expanded capabilities), these predictions range from 0.5 to 1 percentage points while the most optimistic scenario (fast adoption, expanded capabilities) predicts annual labour productivity growth ranging between approximately 0.8 to 1.3 percentage points until 2034.[7] The range of productivity gains of *individual countries* is shaped by both the speed of AI adoption and the technological capabilities of AI systems. *Cross-country* differences are largely influenced by the sectoral composition of each economy. Countries that have a higher concentration of AI-exposed knowledge intensive services are generally likely to have higher productivity gains.

The remainder of this paper is organised as follows. The next section outlines the empirical strategy and provides details on assumptions regarding the micro-level performance gains from AI, the exposure of different sectors to AI and the measurement of current and future AI adoption by businesses. Section 3 describes the quantification results of the different scenarios for macroeconomic productivity gains. Section 4 concludes with a brief policy discussion and potential future extensions.

[6] These reflect the baseline and "AI combined with additional software" estimates from Eloundou et al (2024).

[7] The four scenarios outlined in Samson, Zivkovic and Kalash (2025) touch on a broader range of factors and more qualitative in nature than the quantitative focus of this paper. Nevertheless, their *Flat AI* scenario can be seen as corresponding to the most pessimistic case in the calculations of this paper. Their *United States-centric AI* and *Multipolar AI* scenarios are roughly equivalent for the purposes of this paper's calculations – since it is assumed that all countries in the G7 can *access and use* AI even if AI not *developed* locally – and could be seen closest to the central scenario in this paper. Their fourth, *AGI* scenario, is challenging to quantify in the calculations as it would likely lead to a significant acceleration in the pace of innovation, which is outside the scope of the framework of this paper.

2. AI's productivity effect: going from micro to macro

To estimate sector-level productivity gains over the next decade, this paper adapts the analytical framework proposed by Acemoglu (2024) to a sectoral setting. The empirical implementation of this micro-to-macro framework integrates three key components (Figure 2): (1) **micro-level productivity gains** from AI at the task level, (2) the degree of **exposure of tasks** within sectors to AI, and (3) forecasts of **future AI adoption** across firms within each sector. To the extent possible, the quantification of these components relies on country-specific assumptions.

Formally, projected productivity increases in a given sector *j* of country *c* are derived by multiplying average micro-level productivity gains ($Micro\ Level\ Gains$) with the estimated exposure of sector j in country *c* to AI ($Exposure_{cj}$), and the expected increase in adoption rate of AI ten years into the future ($\Delta\ Adoption\ Rate_{cj[t,t+10]}$):

$$Sectoral\ Productivity\ Growth_{cj,[t,t+10]} = Micro\ Level\ Gains * \ Exposure_{cj} * \ \Delta\ AdoptionRate_{cj[t,t+10]} \tag{1}$$

Given that current adoption rates are not negligible and have shown strong recent dynamics (see Section 2.3 below), some AI driven gains may have already been realised. This paper's calculations focus on the gains that are still to emerge due to the expected *rise* in adoption, in contrast to most previous work.[8]

This framework enables a structured quantification of potential sectoral productivity effects stemming from the diffusion of AI. The subsections below elaborate on the assumptions made to quantify these components across different G7 economies.

2.1. Micro-level performance gains from AI

Most recent studies on aggregate gains from AI rely on evidence on micro-level performance improvements among workers or firms that adopt AI. Following this strategy and reviewing several studies on estimated micro-level gains for *workers* thanks to recent Generative AI, this paper assumes an average baseline effect of 30% on total factor productivity (Figure 3).[9] The micro-level studies are often done in an experimental setting, lending strong credibility to the estimated effects, and cover a range of activities: (i) customer services, (ii) software developers and (iii) professional writing tasks and business consulting tasks. The estimates indicate that the effect of AI tools on worker performance range from 14%, for example in customer service assistance, to 56%, for example in coding.[10] Nevertheless, one may argue

[8] Current adoption rates vary substantially by sector (Annex B, Figure B.4). However, whether differences across sectors will shrink or increase is difficult to predict, due to opposing effects at play. For simplicity, the calculations assume that these differences remain the same, driven by a parallel increase in adoption in all sectors.

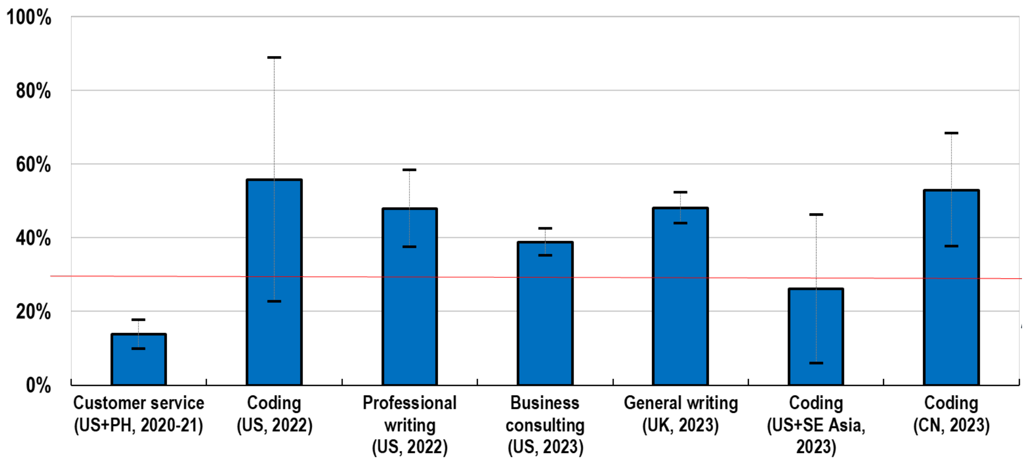
[9] Given that these task-focused studies often estimate time savings, or quantity increases in output when using the same amount of inputs – including capital – this paper interprets them as total factor productivity gains, i.e. lifting the *joint* productivity of labour and capital (computers, office space, etc.). An alternative interpretation of these studies considers that they produce estimates of only labour cost savings (Acemoglu, 2024). This interpretation would mean that in order to arrive at TFP gains, one would need to down-scale these gains to the share of labour, that is to around 60% of the reported productivity gains in the studies.

[10] More specifically, AI support for **customer services** (1st estimate on Figure 3; Brynjolfsson, Li and Raymond, 2025), the effect of AI **coding** assistants on software developers (2nd, 6th and 7th estimates, respectively; Peng et al., 2023; Cui et al, 2024; Gambacorta et al, 2024), and the gains from using Large Language Models such as ChatGPT in speed and quality of **professional writing tasks** (3rd estimate; Noy and Zhang, 2023) or in **business consulting** performances (4th estimate; Dell'Acqua et al., 2023) and the performance improvement with Large Language Models assisting with **general writing** (5th estimate; Haslberger, Gingrich and Bhatia, 2023). Yet more recent studies focus

that these studies were carried out in the context of tasks where performance gains are most promising and may not extend to other business contexts and when AI is used at scale in real-life environments.[11] To remain conservative, the 30% number on micro-level gains that is retained for this paper's calculations averages only the three most precise estimates – excluding the largest effects that are mostly found among programmers.

Figure 3. Micro-level gains from Generative AI are found to be large in a range of tasks

Estimated % increase in productivity and 95% standard errors, as reported by various studies



Note: The graph shows the worker-level productivity effects reported in different studies together with 95% confidence intervals. In parentheses, the reference country and year of the studies are shown.
Source: Compilation from the literature by Filippucci, Gal and Schief (2024).

2.2. *The exposure of different sectors to AI*

AI exposure captures the degree to which AI can potentially affect a specific occupation or sector (Felten et al, 2021; Eloundou et al, 2024). It is derived from the type of tasks that humans carry out and the extent to which AI can do those tasks or helping with them. High exposure of a sector (or occupation) means that it consists of a large share of tasks that AI can assist with - as is the case with knowledge intensive services that comprise of several cognitive tasks. This concept is useful in the process of aggregating micro-level (task-level) performance gains to sector or macro-level productivity effects.

This paper relies on the study of Eloundou et al. (2024), who evaluate which tasks can be performed substantially faster with the help of Generative AI large language models (LLMs).[12] In line with the authors'

on legal services and find very large effects between 34% and 140% (Schwarcz et al, 2025). Briggs and Kodnani (2023) rely on *firm*-level studies which estimate an average gain of about 2.6% additional annual growth in workers' productivity. See Annex B for a detailed overview of considered surveys and harmonisation efforts of this paper,

[11] For arguments that pose risks on the upside, but may arise when focusing on broader productivity measures than worker-level gains (firms and sectors), see a discussion in Section 2.4.

[12] Acemoglu (2024) and several papers building on it follow the same source for quantifying the exposure to AI. An alternative source is due to Felten et al (2021) and its recent update. The two sources correlate strongly at the sector level (Filippucci, Gal and Schief, 2024), although the interpretation of their measure is less easily applicable since their focus is on *differences* across occupations in exposure, not the *share* of tasks exposed.

assessment, this paper uses their results obtained by human evaluators, in particular the following two metrics:[13]

- **Baseline exposure**: The share of tasks for which the required time for completion substantially decreases when using LLMs, relying on the median estimate from Eloundou et al. (2024);
- **Expanded capabilities**: An alternative measure of exposure which also includes tasks where gains are achievable if complementary software is developed *on top of* current LLMs. Essentially, this measure captures AI's expanded capabilities when it is combined with other digital tools.[14]

To aggregate tasks to the sector level, two steps are applied. First, *task*-level AI exposure is mapped into specific *occupations* following the ONET database in the United States. Subsequently, AI exposure of sectors in different countries is derived based on the occupational composition of sectors in each country (see Annex A).[15]

Exposure to AI varies across sectors in G7 economies, with knowledge-intensive services being the most affected (Figure 4). These services rely strongly on cognitive tasks, such as Finance, ICT services (including software development, data services and telecoms), Publishing and Media, and Professional services. In these sectors, between 50% and 80% of tasks are exposed to AI, depending on whether baseline or expanded AI capabilities are assumed. In contrast, the least exposed sectors include sectors with a strong manual, physical task component, such as Agriculture, Mining and Construction. In these sectors, between about 10% and 30% of tasks are exposed to AI. This pattern confirms earlier results on sectoral exposure to non-Generative AI (Felten et al., 2021 and Annex A, Figure A.1), suggesting that Generative AI and LLMs will further expand the effects of previous non-Generative AI. Importantly, the more optimistic scenario with expanded AI abilities shows substantially higher exposure for each sector, while the ranking of sectors is largely preserved.

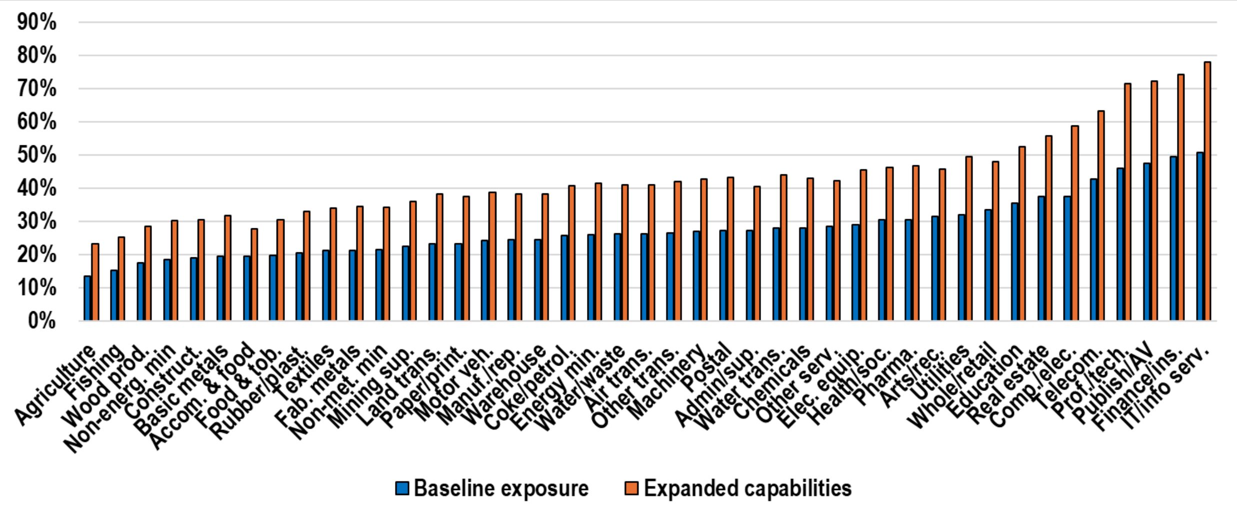
[13] As AI is more deeply integrated with robotics technologies – including manufacturing assembly lines, autonomous vehicles as well as more humanoid robots –, physical, manual intensive tasks will also be more exposed to AI. This is captured by an additional scenario with respect to exposure in Filippucci, Gal and Schief (2024), leading to significant gains in manufacturing and other physical intensive sectors.

[14] The inclusion of this more optimistic and forward-looking case among our scenarios is motivated by the fact that some of the capabilities of LLMs considered by Eloundou et al. (2024) have already improved substantially, such as the length of the context window when interacting with such models as well as their reasoning abilities and accessing recent information online.

[15] Compared to Filippucci, Gal and Schief (2024), where occupational composition of industries in Canada and Japan was assumed equal to the United States, these estimates now leverage detailed data from the Canadian Labour Force Survey to estimate sectoral exposure based on actual occupational composition of industries in Canada, while for Japan the average of the other G7 countries is used for each sector.

Figure 4. AI exposure is highest in knowledge intensive services

The share of tasks in each sector that are exposed by AI, under current and expanded capabilities (average across G7 economies)

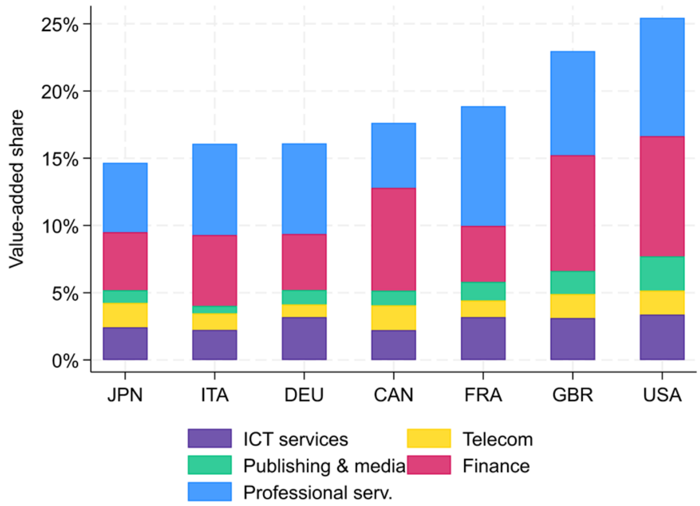


Note: Aggregated from tasks and occupations to sectors, average across G7 economies. Baseline exposure relies on the main task-level measure of Eloundou et al., (2024), while Expanded capabilities rely on the task-level numbers in Eloundou et al. (2024) that are obtained when on AI is combined with additional software.
Source: Authors' calculations based on Eloundou et al., 2024).

Given the important variation in AI exposure across different sectors, the sector composition of G7 economies plays a crucial role in determining to what extent micro-level productivity gains from AI materialise at the aggregate level. The value-added shares of the five most AI exposed sectors – ICT services, Telecom, Publishing and Media, Finance, Professional services – varies substantially across the G7, ranging from 15% to 25% (Figure 5). In the United States, about one quarter of the economy is made by sectors that benefit the most from AI, mostly driven by a large share of Finance and Professional Services, i.e., consulting, legal and accounting, engineering, and science. The United Kingdom shows similarly high figures, around 23%. However, this share is much lower – around 15% – in G7 countries focused more on manufacturing or less knowledge intensive, personal services, such as Japan, Italy, or Germany.

Figure 5. The importance of AI exposed sectors varies across G7 economies

The value-added share of the five most AI exposed sectors in the economy.



Note: When computing the value-added shares, public administration and households as employers are excluded from the calculations.
Source: Sectoral value-added data from the OECD Input-Output tables, 2019.

2.3. Estimating current and future high-intensity AI adoption among firms

In the empirical framework underpinning this paper, the adoption of AI by firms is a key determinant of macroeconomic productivity gains. In practice, measuring adoption is complex, since some firms may use AI in only certain tasks in an ad-hoc manner (e.g. drafting an occasional email with the help of a language model) while others may rely on fully integrated AI systems in their core business activities (e.g. using language models to carry out legal analysis; or assist in creative visual work). Some statistical agencies include questions in their official surveys on several types of AI use. Their figures reveal substantially lower than overall adoption rates when AI adoption in core business operations – an indicator of high intensity use – is considered: for instance, in 2024, the respective figures are about 3% and 14% in Europe, 4 and 9% for the United States and 6 and 8% for Canada (Annex B, Figure B.1).

To remain conservative, and not to overstate the degree of AI use among firms, the calculations focus on this more narrow, high-intensity use in core business functions. However, the accurate identification of such AI adoption faces significant data challenges, in light of the lack of sufficiently comparable data on current AI adoption across countries.[16] This section outlines some important harmonisation and appropriate adjustment steps (sub-section 2.3.1). The empirical strategy to extrapolate our *current* preferred AI adoption to the *future* for a time horizon of 10 years is explained in sub-section 2.3.2.

2.3.1. Current high-intensity AI adoption in core business functions: aiming for cross-country comparability

The lack of harmonised AI adoption data – especially one that also captures the *intensity* of AI use in core business functions – primarily refers to differences in official statistics regarding (i) the purpose of AI use, (ii) the time period the statistics refer to, (iii) firm size coverage, (iv) industry scope, and (v) the survey design, i.e., the definition of a statistical unit and exemplary use cases of AI. To address these discrepancies, we harmonise existing surveys that are sufficiently comparable in measuring high intensity,

[16] Other OECD work on AI adoption has focused on stylised facts that emerge irrespective of the exact definition (Calvino and Fontanelli, 2023).

regular AI use by firms in core business functions (production and service provision) and generate out-of-sample predictions for countries where such adoption rates are unavailable or highly incomparable with the majority of G7 countries (see Annex B for a detailed summary of the underlying steps).

This paper defines AI adoption as the integration of AI in the production process of goods or services, which constitute core business functions. This goes beyond the occasional reliance on AI tools by employees in isolated tasks that are not directly related to core business activities. The first step in harmonising data on AI adoption therefore consists in identifying surveys across different countries that capture the use of AI in production processes of goods or services. Such data collection is carried out by the statistical institutes in Canada, the United States and Eurostat with the latter covering France, Germany and Italy among the G7 (Table 1). As these business surveys differ by their coverage in terms of firm size as well as industry, further data harmonisation consists in excluding micro firms (with less than 10 employees) and to impute data for sectors that are not covered in certain countries (e.g. Agriculture in Eurostat) based on observed differences across sectors in countries where data is available.

Table 1. AI use: When and for what purpose?

Available measures of AI adoption in national statistics

	Low intensity / ad-hoc use of AI in any business function	*High intensity / regular use of AI in core business functions*
Canada	Last year, for any purpose	Last year, in *production*
European Union including G7 countries France, Germany, Italy	Currently, for any purpose	Currently, in *production*
Japan	Past three years*, for any purpose	Past three years*, to improve/introduce goods/services
United Kingdom	Currently, for any purpose	Currently, to *improve operations***
United States	Last six months, in *production*	Last two weeks, in *production*

Note: Adoption intensity captures either more *frequent / more recent* use or integration in production and core business processes. For more details, see Annex B, in particular Table B.4. and its discussion.
*Referring to 2019-2021.
Source: For Canada: Statistics Canada (2024 a, b) - SDTIU (low intensity) and CSBC (high intensity); for EU: Eurostat (2025) Community survey on ICT usage and e-commerce in enterprises; for Japan: National Innovation Survey (J-NIS, 2022); for the United Kingdom: Office for National Statistics (ONS, 2025) - Business Insights and Conditions Survey (BICS); for the United States: United States Census (2025) - Business Trends and Outlook Survey (BTOS).

Despite several harmonisation steps, there are a number of remaining discrepancies between these surveys – especially linked to the time reference and design of surveys. Regarding the former, the time periods referenced in AI usage surveys differ. For example, in the United States, official statistics surveys typically focus on the past two weeks or six months, whereas Eurostat surveys refer to the 'current situation'. Second, while national statistical institutes define a statistical unit of their surveys at least by firm size and economic activity - as it is the case in France and Germany - additional variables, such as turnover, can significantly influence findings on AI adoption across industries.[17] Third, despite efforts in the empirical strategy of this paper to identify comparable questions on AI, differences remain in the level of detail provided to respondents regarding specific AI use cases.

An additional strategy to mitigate measurement related differences across countries in AI adoption is to rely on information about its underlying drivers, such as digital infrastructure and skills. [18] Indicators for these drivers are more established and more comparable across countries. To select the most relevant predictors of AI adoption, a regression-based analysis is used. This approach also helps to fill remaining

[17] For instance, while France includes turnover in its definition of a statistical unit, Germany does not. Expecting that high-turnover (frontier) firms exhibit faster and more intensive AI adoption, a sampling strategy that does not consider turnover categories may risk suffering from a selection bias.

[18] This is motivated by previous work that found a crucial role for these factors in enabling and driving digital technology adoption (Nicoletti, Rueden and Andrews, 2020). The exact drivers included in the regression analysis are pinned down by the availability of cross-country comparable indicators and their ability to provide a good fit for capturing AI adoption. Digital infrastructure is measured by the share of individuals using the internet and the number of data centers per capita, while skills are represented by the proportion of STEM graduates (minimum Bachelor's degree) in the population aged 20-64, alongside PIAAC results for problem-solving abilities. Relevant variables were identified using the Lasso methodology, which selects variables that are jointly provide the best prediction for the dependent variable (Annex B, Table B.3).

data gaps for Japan and the United Kingdom where comparable survey data on high intensity AI adoption in core business functions is unavailable.

To leverage the econometric relationship between AI adoption and its drivers across all countries, while further addressing the remaining comparability issues, the preferred high intensity adoption rate estimates in the paper are obtained as the average between the harmonised values and the predicted values. Following this empirical strategy the **estimates of *current* high-intensity AI adoption** by firms in core business functions range between about 2% for Japan and Italy and about 6% for United States and Canada (see also Figure 9 in Section 2.4).[19, 20] While the empirical strategy of this paper significantly increases the comparability of cross-country data, there are still discrepancies that cannot be corrected under the current availability and scope of data. For this reason, an essential step to expand data-based research and monitoring is to design harmonised surveys on AI.

2.3.2. Future AI adoption: following the path of previous GPTs

Evaluating the impact of AI over the next 10-years requires making assumptions about the path and speed along which the technology is adopted by firms. For this purpose, previous major GPTs are considered such as electricity, ICT tools including computers and the use of the internet as well as mobile phones. Historically, GPTs have demonstrated S-shaped adoption curves, which typically reflects the gradual uptake of new technologies, followed by rapid adoption and then a plateau as saturation occurs (see the example of the internet in Annex B, Figure B.6 and Hall, 2009; Geroski, 2000; Tankwa et al, 2025, building on Griliches, 1957, and Rogers, 1962).

In the United States, key past GPTs such as electricity and Information and Communication Technology (ICT) tools (including computers and the internet) reached adoption levels of 23% and 40%, respectively, 10 years after the user-friendly breakthrough of these technologies (Figure 6). Notably, adoption rates are higher for more recent technologies, with mobile phones reaching 60% adoption rate.[21] These figures highlight the accelerating pace of adoption for newer technologies compared to earlier innovations, as was discussed in previous work (Comin and Mestieri, 2018; Tankwa et al, 2025, see Annex B, Figure B.5).

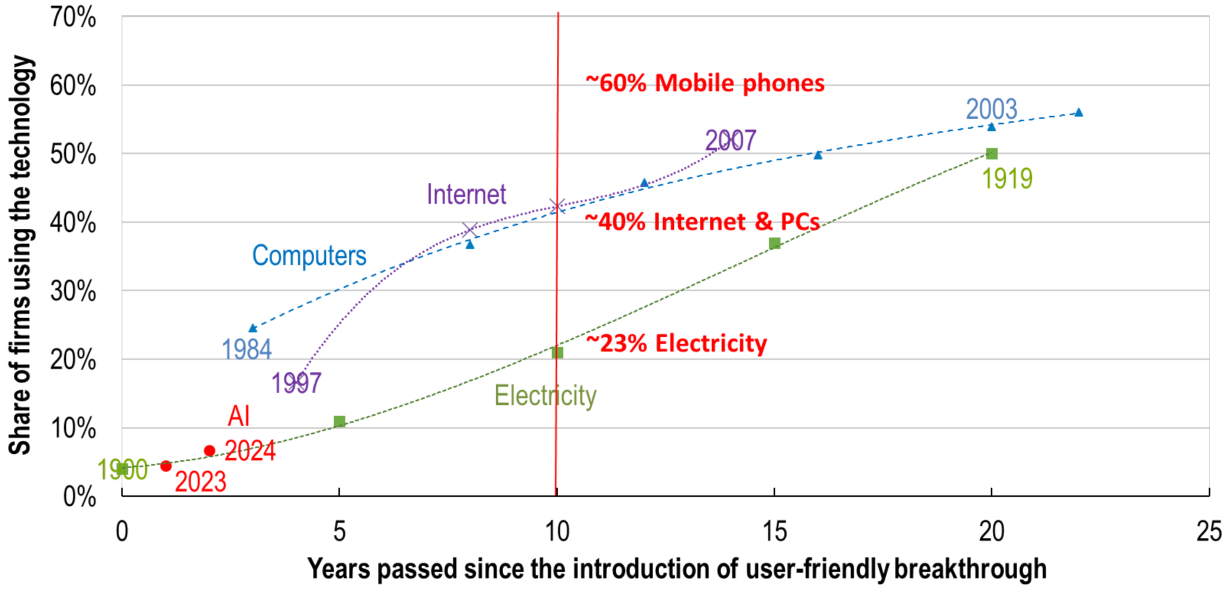
[19] The resulting estimates for high-intensity, core business function use AI adoption rates among firms are as follows, from highest to lowest: United States (6.1%), Canada (5.7%), United Kingdom (5.4%), Germany (4.8%), France (3.1%), Italy (2.2%) and Japan (1.9%).

[20] Several ad-hoc cross-country surveys on AI use are consistent with the relative position of the United Kingdom and Japan. For instance, the United Kingdom's position behind the United States and Canada aligns with findings of a recent survey of the OECD's Global Forum on Productivity which also confirms relatively low levels of AI adoption in Japan. Additionally, a survey by Japan's Ministry of Internal Affairs and Communications indicates that the regular use of AI in Japanese businesses is approximately one-third of the level seen in the United States, a proportion that aligns with the harmonised and predicted values of this paper (see Annex B, Figure B.3).

[21] For mobile phones, adoption rates reflect individuals instead of firms due to the lack of data.

Figure 6. Mapping AI's future adoption path with that of previous General Purpose Technologies

Share of firms* using the technology, in the United States



*For mobile phones, the share of individuals is shown due to the lack of data.
Source: For computers and internet: United States Census; for electricity adoption of businesses: Woolf (1987) and Our World in Data; for mobile phone use of individuals: International Telecommunication Union; for AI: United States Census Bureau, Business Trends and Outlook Survey (BTOS). Although this figure highlights the S-shaped adoption patterns of past GPTs in the United States, comparable trends are also observable more broadly, such as in internet usage across G7 economies (see Annex B, Figure B. 6).

Based on historical evidence of the adoption of these previous GPTs, this analysis outlines **three AI adoption scenarios at low, medium, and fast pace** that align with the respective adoption paths of electricity, computers and the internet, and a more recent digital technology, mobile phones.

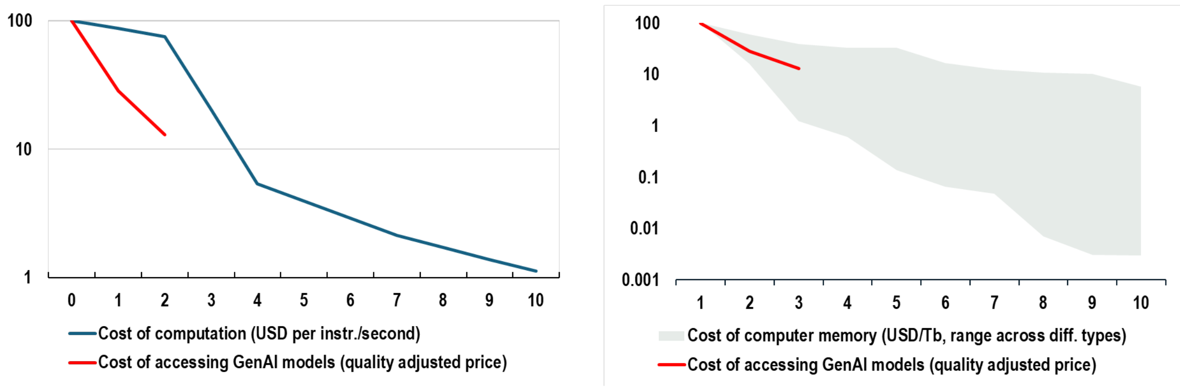
The fast-evolving patterns of recent increases in AI adoption and declines in the cost of accessing AI are suggestive of a faster adoption trajectory. For instance, AI adoption increased by about 50% from one year to the next both among EU countries and in the United States in official data, from initial rates of around 3-5%.[22] Such an initial steep rise is most consistent with the S-shaped adoption pattern which was obtained for more recent digital technologies (mobile phones). Assuming the continuation along such S-shape paths lead to an adoption rate of about 60% in 10 years. In addition, new evidence shows that the quality-adjusted cost of AI models is declining exponentially (about 80% over the past two years; Andre et al, 2025), mirroring past trends observed in computational costs and computer memory (Figure 7). Coupled with Generative AI's user-friendly nature, which is mostly based on human language interaction, its improving cost-effectiveness lends further support for a more dynamic future adoption path. On the other hand, a deeper integration into core business functions may still require significant complementary investments in terms of data, skills and a reorganisation and rethinking of business processes (Brynjolfsson, Rock, Syverson, 2021).

[22] For EU: 2023 to 2024; for United States; 2024 to 2025 (Eurostat, 2025; United States Census Bureau, 2025).

Figure 7. AI cost decline seems comparable to what happened with other digital technologies in the past

A. Historical costs of computation vs. cost of AI
Index, first year of data =100

B. Historical costs of computer memory vs. cost of AI
Index, first year of data=100



Source: Cost of computation: Kurzweil (2024); Cost of computer memory: McCallum (2023). Cost of accessing GenAI models: Andre et al. (2025).

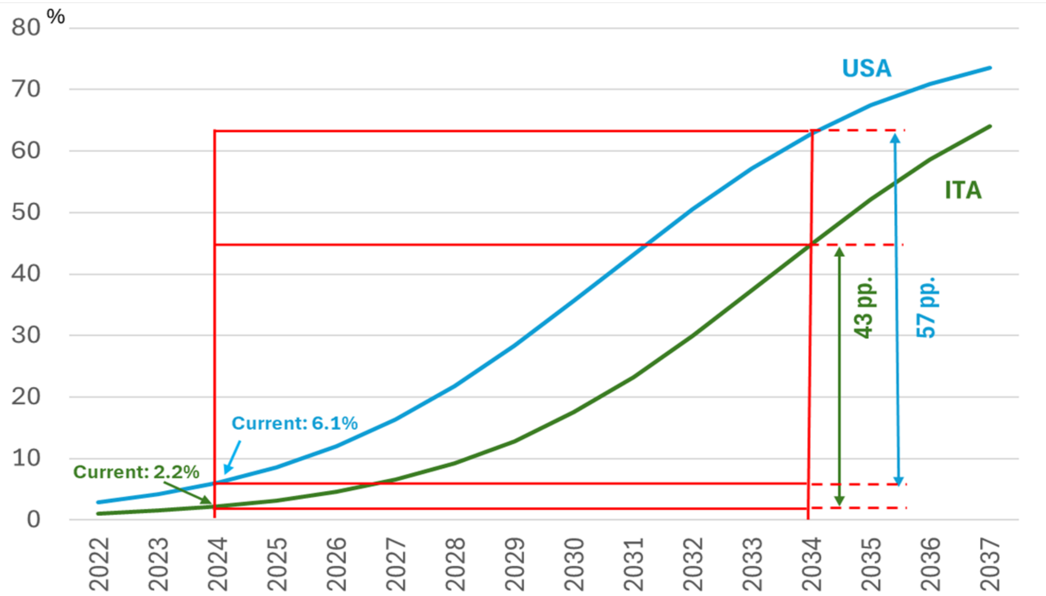
Future AI adoption patterns in the paper are approximated by an S-shaped function that is defined by parameters on the exact shape and the maximum level, which determine the speed of adoption when considered over any given time horizon.[23] In the calculations of this paper, the curve for each country's AI adoption trajectory is pinned down by the current adoption rates of countries (which vary across countries) and assumptions on these parameters (which are assumed to be the same across countries). This approach gives rise to significant cross-country differences in adoption rates, which reflect both initially faster adoption rate increases (leading to a divergence) followed by a slowdown as countries near a saturation point (leading to convergence).

Over the projection horizon of 10 years-time, it turns out that the first impact dominates, leading to wider difference in adoption than currently observed, as illustrated by Figure 8 with the example of the United States (early adopter) and Italy (lagging adopter). The United States and Italy have AI adoption levels of about 6% and 2% respectively, in 2024, hence their increase in AI adoption along the S-shaped path over the next 10 years differs substantially (Figure 8 illustrates this under the fast adoption scenario, mimicking the diffusion of mobile phones). Accordingly, within 10 years AI adoption in the United States is expected to increase by 57 percentage points while AI adoption in Italy is expected to increase by 43 percentage points. This divergence is due to the fact that both countries are still on the accelerating parts of their adoption paths, during which level difference in AI adoption across countries tend to increase.

[23] Specifically, AI adoption paths are modelled by fitting logistic curves to current AI adoption rates, under different assumptions on the logistic growth rate (the "speed of adoption") and assuming a maximally achievable adoption rate of 80% in the long run. For example, for the rapid adoption scenario, the adoption path is modelled by fitting a logistic function with a logistic growth parameter similar to that observed for mobile phones (see Figure A.5 in Annex B).

Figure 8. Current adoption rate differences across countries likely to drive adoption 10 years from now: United States versus Italy

Future AI adoption path assumptions following the S-shape of previous GPTs under fast adoption speed scenario (following mobile phones), % of businesses



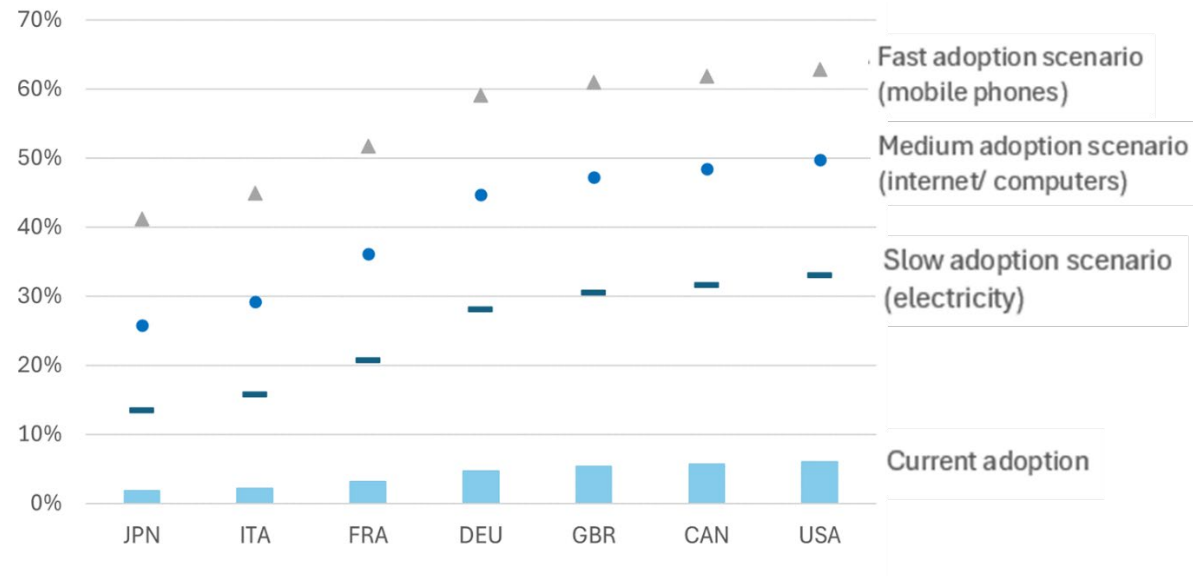
Note: Calculations based on the adoption speed of the latest digital technology (mobile phones). Adoption speed is sourced from Tankwa et al. (2025). See Annex B, Figure B.5 for more details.

By applying the three AI adoption scenarios assuming a slow, medium or fast adoption speed aligned with the adoption speed of previous technologies electricity, ICT, and mobile phones, our estimations project that AI adoption levels will vary significantly across countries by 2034. More specifically, AI adoption in 2034 is expected to range from 30% to about 60% for the United States, Canada, the United Kingdom and Germany, while lower AI adoption levels are predicted for France, Italy and Japan, between 15% to 50% (Figure 9). These differences across countries in terms of future assumed adoption rates reflect the ranking of estimates for current AI adoption levels – in which the United States, Canada and the United Kingdom, for example, show highest AI adoption across countries (see sub-section 2.3.1.).

The uncertainties around these projections are considerable. On the on hand, countries with low current level of AI adoption may be able to learn from the experience of other countries that have already achieved higher levels of AI adoption, allowing them to achieve a steeper adoption path compared to the leading countries (“catching-up”). On the other hand, future AI adoption may be inhibited by the same frictions that explain lower AI adoption rates today, so that laggard countries will experience flatter adoption paths. This seems to have been the case for internet use, when comparing the experience across G7 countries (Figure A.6 in Annex B). Given the presence of both upside and downside risks for follower countries, the projections in this paper are based on the assumption that countries differ in the timing but not the shape of the AI adoption paths.

Figure 9. The expected increase in AI adoption varies a lot across countries

Current and future adoption (in 10 years) based on the S-shaped adoption paths seen in previous GPTs, % of businesses



Note: Current adoption rates are taken from official national statistics after harmonisation steps (CAN, EU, United States) or when this is not possible, using predictions as a function of the digital infrastructure, skills and the sectoral composition (JPN, GBR).
Source: Authors' calculations.

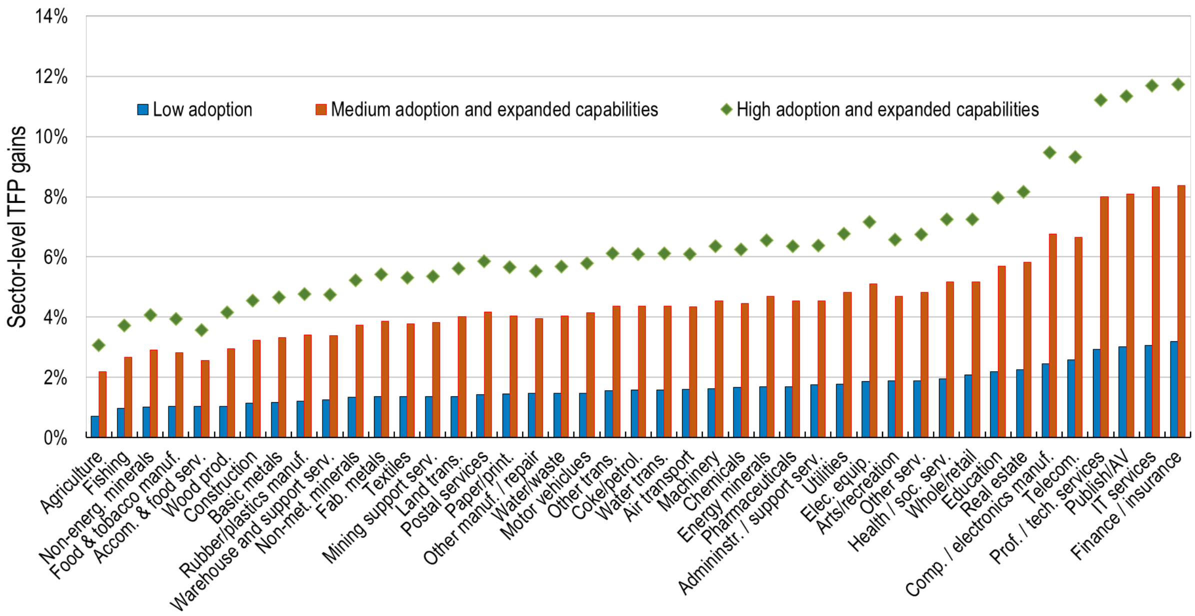
2.4. Predicted sector level productivity gains

Projections of sector-level TFP gains over the next decade are obtained by multiplying estimates of micro-level productivity gains, sectoral exposure to AI, and projected AI adoption rates, following equation (1). Significant differences in projected productivity gains exist across sectors, reflecting the fact that some sectors are much more exposed to AI than others. Depending on the scenario, projected cumulative TFP growth ranges from a low 1% in manual-intensive industries (e.g., Agricultures, Fishing, or Mining) to over 10% in knowledge-intensive services, such as Professional and Technical Services, IT Services, or Finance (Figure 10, showing averages across G7 countries). Across all sectors, the highest projected productivity gains are obtained under the scenario that assumes the development of complementary software and adoption rates similar to those of mobile phones in the past ("rapid adoption and expanded capabilities" scenario).

These projected productivity gains could understate the true gains from AI to the extent that additional productivity gains can result from broader AI-driven innovations in organizational structures and business models. Such gains would not be observed at the level of individual tasks but would emerge in the productive re-configuration of the interlinkages between existing work tasks or in the creation of entirely new tasks. While these possibilities constitute upside risks to the growth projections presented here, there also exist downside risks. Acemoglu (2025) stresses the possibility that AI could lead to the creation of "bad" tasks which generate revenue but have negative social value, such as the design of algorithms to manipulate buyer choices or to create addictive social media. Misuse of AI could also depress measured productivity, for example if value is destroyed through AI-powered malicious computer attacks.

Figure 10. The projected AI-driven productivity gains vary across sectors

Projected sector-level total factor productivity gains under different scenarios



Note: calculations follow the description in Section 2.1, in particular equation 1.
Source: Authors' calculations.

3. AI's expected impact on aggregate productivity across the G7

The second step of the empirical strategy in this paper involves translating sector-level TFP gains into aggregate labour productivity growth (Figure 2). This is achieved by summing the sectoral TFP gains, with each sector weighted by its *current* share of value added in aggregate GDP. This approach is grounded in Hulten's theorem and constitutes a first-order approximation of the full effect of AI, which would also reflect changes in sectoral output prices and the reallocation of factors of production across sectors.[24] The resulting estimate of the aggregate TFP gain is then converted into labour productivity growth by applying a multiplication factor of 1.5, which accounts for the role of capital deepening.[25]

The aggregation approach is most accurate when the sectoral composition of the economy remains relatively unaffected by differences in sectoral productivity growth. Thus, a notable phenomenon that is not captured by this aggregation approach is Baumol's growth disease (Nordhaus, 2008), which can dampen aggregate GDP growth if sectors with slower productivity growth expand as a share of GDP, as has been the case historically with the rise of services in most economies (Box 1). The possibility that aggregate productivity gains could be limited because AI increases productivity only in the most AI-exposed sectors

[24] This theorem, which applies in efficient economies irrespective of their underlying structure, is a foundational result in productivity analysis that is often invoked to assess the aggregate impact of micro-level productivity shocks (Hulten, 1978; Acemoglu, 2024).

[25] A multiplier of 1.5 consistent with standard growth models with a Cobb Douglas production function and a capital and labour share of 1/3 and 2/3, respectively. It also matches the ratio of labour productivity growth to TFP growth historically, in particular over the past five decades in the United States (Bergeaud, Cette and Lecat, 2016).

while not benefitting all other ones – personal services or manual task intensive sectors – was also stressed by Aghion, Jones and Jones (2017).

Filippucci, Gal, and Schief (2024) employ a multi-sector general equilibrium model to assess the role of Baumol's growth disease in shaping the aggregate gains from AI over the next decade. Their findings show a relatively limited role of this channel in the main scenarios, even with unevenness in sectoral productivity gains from AI, provided that demand is sufficiently reactive to prices and factors of production can efficiently be redeployed across sectors. Moreover, cross-country differences that would arise due to the Baumol effect are difficult to pin down and interpret, given uncertainties in measuring and mapping structural rigidities into model parameters. Hence this paper relies on the simple aggregation strategy described above and discusses the Baumol effect only in general terms in Box 1.

Box 1. AI, sectoral reallocation and the Baumol effect: implications for aggregate productivity growth

Historically, sectors experiencing lower productivity growth have tended to see increases in both their employment and GDP shares, as has been the case for services in most advanced economies. Over the long run, this pattern has contributed to lower aggregate productivity growth, commonly referred to as **Baumol's growth disease** (Nordhaus, 2008). Given that the most productive applications of AI are limited to a few sectors (Figure 10), AI could induce shifts in the sectoral composition of the economy that increase the GDP share of the least AI-affected sectors. Put differently, as Aghion, Jones and Jones (2017) argue, overall growth from AI may in the longer term be limited "not by what we do well but rather by what is essential and yet hard to improve".

Filippucci, Gal and Schief (2024) analyse under what conditions a significant Baumol effect can arise that would undermine AI-driven productivity growth during the next decade. They find that while sectoral differences in productivity growth are the fundamental drivers of the Baumol effect, its magnitude also depends on *demand-side factors* and the flexibility of *factor reallocation*.

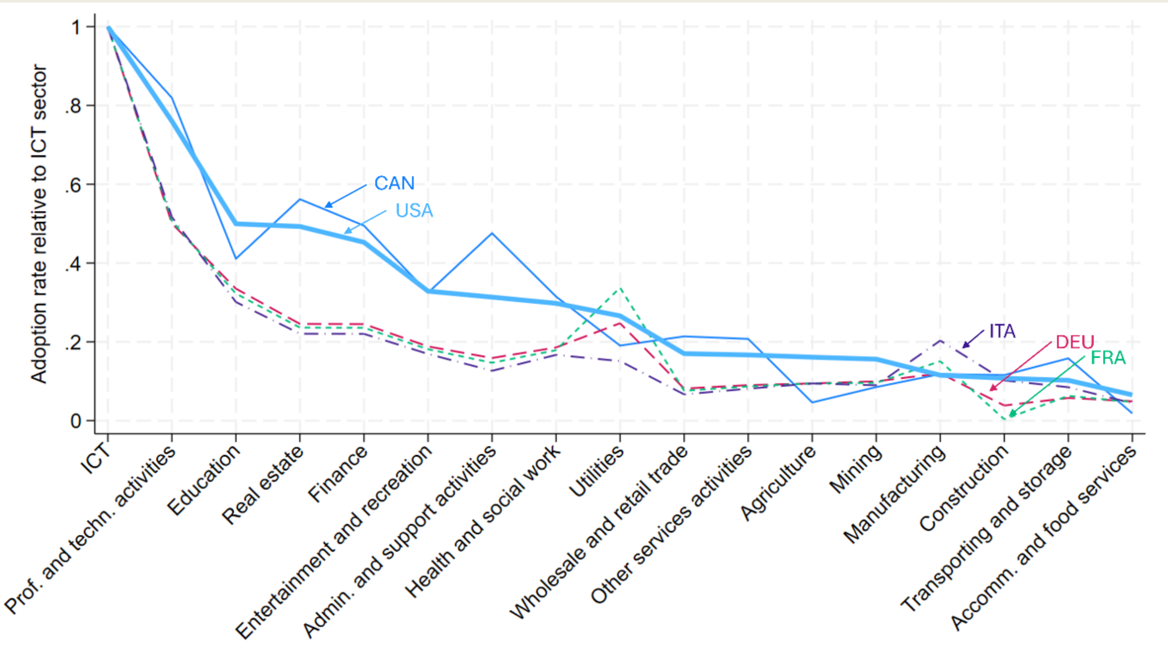
- On the *demand side*, if firms and consumers are highly responsive to changes in relative sectoral prices - readily substituting goods and services from one sector for those of another when prices shift - the Baumol effect will be mitigated. Conversely, if consumers exhibit strong preferences for maintaining their consumption of goods and services from low- and high-growth sectors in relatively fixed proportions, the GDP share of low-growth sectors will expand, thereby exerting downward pressure on aggregate productivity growth. This could be the case, for example, if there is limited demand for additional output from strongly AI-impacted sectors, such as legal services or finance.
- *Factor mobility* across sectors further influences the size of the Baumol effect, particularly when demand is price inelastic. When productive resources can be easily reallocated across sectors, strong productivity gains in AI-intensive industries can be leveraged by redeploying factors of production to those sectors that do not experience direct productivity improvement but produce goods and services that consumers value. However, if factor mobility is constrained, sectors with limited AI adoption may fail to expand their output, as they neither benefit from AI-driven productivity gains nor attract an inflow of productive resources. As a result, their relative prices will rise substantially, increasing their GDP share and suppressing aggregate productivity growth.

Quantifying the role of these assumptions, Filippucci, Gal, and Schief (2024) conclude that the Baumol effect is unlikely to reduce aggregate productivity growth by more than one-sixth over the next decade in most scenarios. However, it could have a more significant impact under extreme scenarios that assume highly uneven sectoral gains, price-inelastic demand, and limited factor mobility.

To give some indication about country-level differences in the drivers of the Baumol effect, adoption rate differences across sectors are insightful: more concentrated AI adoption across sectors will lead to more uneven sectoral productivity gains, which could act as a drag on aggregate productivity. In every G7 economy, the leading sector in terms of AI adoption is the ICT sector. However, comparing the AI adoption rates in other sectors with that in the ICT sector of the same country reveals that AI adoption is more widespread in the United States and Canada compared to France, Germany, and Italy (Figure 11).

The relatively low AI adoption rates in sectors such as Finance or Wholesale and Retail in France, Germany, and Italy depresses the aggregate productivity gains from AI in these countries. Most importantly, it reduces the overall AI adoption rate in the economy, which could be higher if the AI adoption gap in lagging sectors vis-à-vis ICT was less pronounced, as observed in the United States and Canada. Moreover, it could also depress the aggregate gains from AI indirectly to the extent that the implied uneven sectoral productivity growth across sectors gives rise to a negative Baumol effect.

Figure 11. AI adoption appears more widespread across sectors in the United States and Canada than in other G7 countries



Source: Authors' calculations relying on the preferred harmonised AI adoption values (see calculations in Section 2.3.1).

3.1. *Aggregate productivity from AI over the next decade*

Predictions on the contribution of AI to annual labour productivity over the 10-year time horizon differ significantly across G7 economies and scenarios (Table 2 and Figure 12). Across countries, these range from 0.2 to 0.4 percentage points under the slow adoption scenario, from 0.5 to 1 percentage points under the medium adoption and expanded capabilities scenario and from 0.8 to 1.3 percentage points under the rapid adoption and expanded capabilities scenario.

Table 2 Expected aggregate productivity gains from AI across G7 economies

Scenario	Exposure given AI capabilities*	AI adoption pace**	AI's predicted contribution to annual labour productivity growth over the next decade (in p.p.)						
			USA	GBR	DEU	CAN	FRA	ITA	JPN
Slow adoption	Baseline	Slow (as electricity)	0.41	0.39	0.34	0.35	0.26	0.19	0.16
Medium adoption and expanded AI capabilities	Expanded	Medium (as computers & internet)	0.99	0.97	0.86	0.86	0.72	0.57	0.51
Rapid adoption and expanded AI capabilities	Expanded	Rapid (as mobile phones)	1.28	1.27	1.16	1.13	1.05	0.89	0.82

*Exposure to AI is measured as the weighted share of tasks in which AI can substantially reduce the time required for their completion, constructed as outlined in Annex A. Baseline exposure refers to the median estimate of task-level exposure in Eloundou et al. (2024). High exposure refers to the upper-end estimate of task-level exposure in Eloundou et al. (2024), which makes more optimistic assumption about the integration of AI via the development of complementary software (see Figure 4). AI exposure can vary across countries due to differences in the occupational structure within sectors and the sectoral composition of the economy.
**The pace of AI adoption is benchmarked to that of previous technologies (see Figure 6).

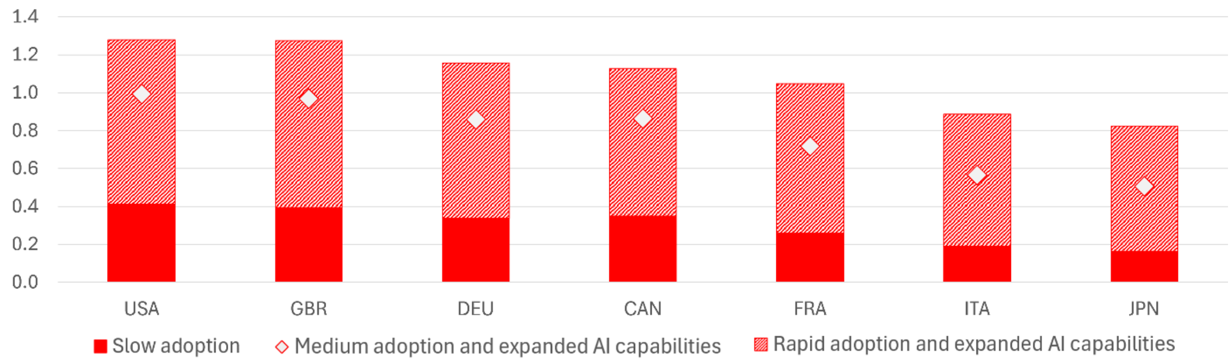
These productivity gains differ significantly across G7 economies reflecting both the pace of AI adoption and the sectoral composition of national economies. For instance, estimates are comparably low in Japan and Italy, around 0.2-0.8 across scenarios, consistent with having a lower share of AI-driven knowledge-intensive service sectors and a larger emphasis on manufacturing which is less impacted by current AI. In contrast, the United Kingdom and the United States, with a greater reliance on AI-impacted sectors, show productivity gains from 0.4 to 1.3 percentage points.[26]

The high end of the range of these estimates are higher than the previous main scenario in Filippucci, Gal and Schief (2024), driven by a dynamic increase in adoption rates across both sides of the Atlantic, and the rapid fall of quality adjusted prices of accessing AI models. These patterns serve the basis for considering a more rapid speed of adoption scenario going forward (matching that of mobile phones) than in previous work (matching that of the internet and PCs).

[26] Filippucci, Gal and Schief (2024) consider an additional scenario where AI is combined with robotics technologies but which is more difficult to quantify in the framework. Nevertheless, the main conclusion is that stronger gains can be expected in manufacturing and other physical intensive sectors, thus raising expected aggregate gains relatively more in economies that specialise in these activities.

Figure 12. AI's macroeconomic productivity gains can be significant but adoption and sectoral specialisation are key

Predicted labour productivity growth due to AI over the next 10 years, percentage points (annualised)



Note: the same notes apply as below Table 2.
Source: Authors' calculations.

3.2. *Implications for current productivity developments due to AI*

The estimates in Figure 15 show significant impacts of AI on future productivity over the medium term (10-year horizon), However, this does not necessarily imply that AI-driven gains will already appear in official productivity statistics in the short run. The estimates are based on comparing two equilibrium states and do not consider adjustment costs in adopting AI that can occur in the short run in between. For example, when technologies – such as computers and the internet – require significant complementary investments in both hardware and intangibles (in additional software, data or firm organisation), measured productivity growth may respond slowly to adoption, or even show a temporary decline.[27]

This phenomenon, known as *J*-curve (Brynjolfsson, Rock and Syverson, 2021), yields an initial under-estimation of productivity growth, and-over estimation in the following periods. This may very well apply to AI as well, despite its more user friendly nature than previous technologies. The reason is that a full integration of AI in business processes may still require a rethinking of business processes, experimentation and additional investments in intangibles such as data and skills.

Assessing AI's role in current productivity developments is thus challenging to do in real time. This is further complicated by the fact that relevant productivity measures – sectoral TFP – usually come with a significant lag, and reflect the impact of a series of other economic developments beyond AI.

An alternative way to track the importance of AI and gauge its potential productivity effect in the short run is to closely monitor AI adoption, also by sectors (see Annex B, Figure B.4). To do that in a cross-country comparable manner, however, statistical surveys on AI use should be substantially more harmonised, as explained above (Section 2.3 and Annex B).

[27] This occurs due to the under-estimation of intangible investments.

4. Concluding remarks and future extensions

This paper evaluates the potential gains in productivity from AI in G7 economies over a 10-year horizon. Building on Acemoglu (2024) and updating Filippucci, Gal and Schief (2024), it combines existing estimates of micro-level performance gains with evidence on the exposure of activities to AI and likely future adoption rates. In particular, it provides more refined and updated estimates for all individual G7 countries beyond the United States, based on harmonised measures of high intensity AI adoption in core business functions, ranging between about 2% and 6%. Importantly, future AI adoption is derived from insights gained on previous GPTs characterized by an S-shaped curve at different adoption speeds – slow, medium and fast.

Results show that annual labour productivity gains from AI are likely to differ significantly across G7 economies. Under different scenarios this paper estimates annual labour productivity gains over the next decade ranging between 0.2 and about 0.8 percentage points in Japan and Italy, whereas annual gains for the United Kingdom and the United States range between 0.4 and 1.3 percentage points. These differences are driven by several factors. For example, the range of productivity gains in individual countries is influenced by both the speed of AI adoption and the capabilities of AI technologies, while cross-country variations are largely attributable to the sectoral composition of economies, with countries that have a higher share of AI-exposed industries typically experiencing greater impacts.

Harvesting aggregate gains from AI is based on the effective use and adoption of AI by businesses and access to the globally most advanced AI models. Investing in digital infrastructure for AI - starting with wide-ranging internet access for companies -, strengthening AI-related skills including in the field of STEM sciences and ensuring healthy competition are essential (OECD, 2023c; Andre et al.,2025). Moreover, in view of often uneven productivity growth across sectors, policymakers should also aim for easing the movement of labour and capital across sectors, which may otherwise limit the overall economic benefits. In this context, enhancing retraining programs for workers and ensuring the effective functioning of capital markets are crucial steps.

This paper is a first step in the extension of Filippucci, Gal and Schief (2024). Future research on the productivity impact of AI will further extend the scope of considered countries beyond G7 economies and zoom in on the role of international linkages in spreading AI gains across countries. *A priori,* the role of international openness in shaping AI gains can be manifold: first, AI in the context of global value chains where gains from AI are transmitted by trading intermediates; second, the potential of AI to alter global trade by reducing trade barriers (due to language differences) and potentially changing comparative advantages of countries (re-shoring); third, via trade in digital services, to ensure the accessibility of frontier AI models in non-AI developer economies.

Beyond the research on the impact of AI on competition, or – as discussed in this paper – productivity, the OECD is also making efforts to improve the availability of AI-specific data. Initiatives such as the ongoing work in the OECD's AI Observatory, with data collection on AI infrastructure, or specific projects with a focus on adoption and their impacts on firms in selected countries are important elements in this regard (OECD, 2025; Russo et al., forthcoming; Calvino and Fontanelli, 2023). To implement and further expand this research agenda to inform effective policy making, as well as to monitor the diffusion of AI, a key step forward would be the harmonisation of data on AI adoption.

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Annex A. Calculating sector-level exposure to AI in different countries

To derive sectoral AI exposure in terms of ISIC sectors, we start from AI exposure estimates from Eloundou et al. (2024) at the level of ONET tasks.[28] We use both their baseline beta measure and the gamma measure, which accounts for extended capabilities. Eloundou et al. (2024) offer two types of exposure estimates: those derived from human ratings of ONET's Detailed Work Activities (DWAs) and task-level assessments using GPT-4 APIs. We focus on human-coded exposure variables. While the GPT-based measures offer more granular insights, Eloundou et al. (2024) present the human-coded estimates as their baseline findings, arguing that the GPT-based results are less robust and more sensitive to prompt variations. We combine this exposure data with task-level information on manual task intensity, sourced from Acemoglu and Autor (2011).

For the United States, we aggregate the task-level data to the occupational classification of SOC 2018 at the 6-digit detail. We differentiate between core and non-core, supplemental tasks in occupations and assign a double weight to core tasks (see more details in O*NET, 2023). Using the occupational composition of industries from the BLS Occupational Employment and Wage Statistics (OEWS) survey, we aggregate this information to NAICS 3-digit industries using employment weights. We then convert the NAICS 3-digit classifications to ISIC 2-digit sectors by leveraging a crosswalk from the U.S. Census Bureau, corresponding NAICS 6-digit to ISIC 4-digit industries, incorporating employment data at the NAICS 3-digit level. In cases where a single NAICS 6-digit code corresponds to multiple ISIC 4-digit codes, we first distribute the employment evenly across the NAICS 6-digit codes within each NAICS 3-digit group, then again equally across each ISIC 4-digit code corresponding to the same NAICS 6-digit code, generating unique NAICS 6-digit–ISIC 4-digit cells with their associated employment estimates. Finally, we aggregate the AI exposure to the NAICS 3-digit level and convert it to ISIC sectors using an employment-weighted average across the NAICS 6-digit–ISIC 4-digit cells.

For EU countries and United Kingdom, we first aggregate the task-level dataset to the 8-digit O*NET-SOC 2018 occupation codes, again using core weights. We then convert these codes to SOC 2010 6-digit codes, for which a crosswalk to ISCO 2008 4-digit codes is available from the BLS. To avoid double-counting, we follow the methodology outlined by Dingel and Neiman (2020). When a SOC 2010 6-digit code maps to multiple ISCO 4-digit codes, we allocate the U.S. employment weight of the SOC across the ISCO codes proportionally to the ISCO employment shares in the respective EU country. This gives us unique SOC 2010 6-digit–ISCO 2008 4-digit cells. We then calculate the average AI exposure across all SOCs within each ISCO 3-digit category, weighted by estimated employment in the SOC 2010 6-digit–ISCO 2008 4-digit cells. Finally, we use an ad-hoc extraction of Eurostat microdata providing the composition of ISCO 3-digit occupations within ISIC 2-digit industries (with some aggregation to avoid overly granular cells with too few units) to obtain a weighted average of AI exposure within ISIC sectors.

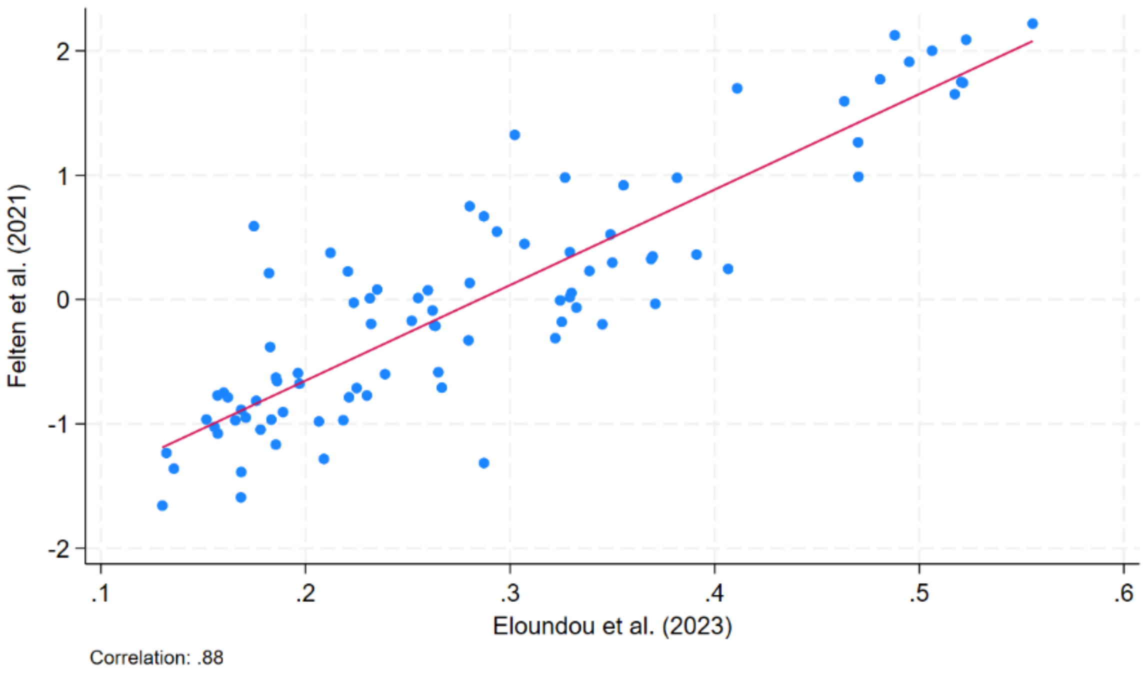
[28] The implicit assumption is that these task-level estimates apply in other contexts as well outside the United States. Given the high granularity and detail at which the tasks are assessed, and that the group of countries included in the sample are all highly developed ones (G7 economies), this seems reasonable. Further work can test this assumption, provided that country specific exposure measures become available.

For Canada, we convert exposure for United States occupations at SOC 2010 to SOC 2018. Then, we use a crosswalk provided by Statistics Canada to convert our estimates to Canadian NOC 2016 classification, and then to NOC 2021. Two assumptions are required in this passage: that workers in each NOC 3-D are distributed following the relative employment weight of each United States SOC 6-D category with respect to the corresponding Canadian NOC 3-D, and that workers are uniformly distributed workers in multiple mappings. We then collapse to the more aggregated NOC 43 categories, for which industry by occupation distribution of employment is available in LFS Canada. As Canada uses a similar NAICS classification for industries, we follow the same procedure as United States to convert NAICS into ISIC sectors.

Finally, for Japan the same occupational composition in each sector as in the average of G7 is assumed, due the lack of data.

Figure A.1. High correlation between alternative AI exposure estimates at the occupation level

Felten et al. (2021, non-Generative AI) and Eloundou et al. (2024, Generative AI)



Note: Correlation at the United States-SOC occupation level.
Source: Authors' calculations using the cited sources.

Annex B. AI adoption calculations

Current high-intensity AI adoption in core business functions: aiming for cross-country comparability

The paper estimates high-intensity AI adoption in core business functions by firms in a harmonized manner across countries. It is defined in the paper as the integration and regular use of AI in core business processes, that is in the production process of goods or services. This concept goes beyond the occasional use of AI tools by employees in isolated tasks that are not directly related to the main activities of firms.

Statistical agencies put in place surveys to measure AI use by firms, but sometimes the focus is less precise than such high-intensity AI adoption in core business functions. Hence the paper relies on several steps to estimate it: First, the identification of official country-level business survey questions that aim to capture high intensity AI use in core business functions; second, the harmonisation of the measures derived from these questions, with respect to firm-size and industry coverage; and third, relying on the underlying drivers of AI use – indicators on digital infrastructure and skills – in econometric analysis to estimate structural AI adoption capacity. Indicators about these underlying drivers are more standard and more comparable across countries than AI adoption, hence their use can further improve comparability.

The estimates based on them are then used for two purposes:

1. Obtaining out-of-sample predictions for those countries' data that cannot be included in the harmonisation step due to the lack of sufficiently comparable adoption data (United Kingdom and Japan);
2. Enhancing the comparability of AI adoption measures for the other countries by taking a simple average with the harmonised values to obtain the final preferred estimate for high-intensity AI adoption in core business functions.

Identification of surveys covering high intensity AI use in core business functions

The first step in harmonising data on AI adoption involves identifying official statistical surveys across different countries that capture the use of AI in production processes of goods or services (Table B. 1). [29] Such data collection is carried out by the statistical institutes in Canada, the United States and Eurostat with the latter covering France, Germany and Italy among the G7 (Table B. 1).

[29] Surveys by private institutions and other ad-hoc surveys were also considered but did not seem sufficiently useful to include them in the analysis due to their smaller sample size or inadequate representativeness.

Table B.1. Questions in official statistical business surveys about AI use

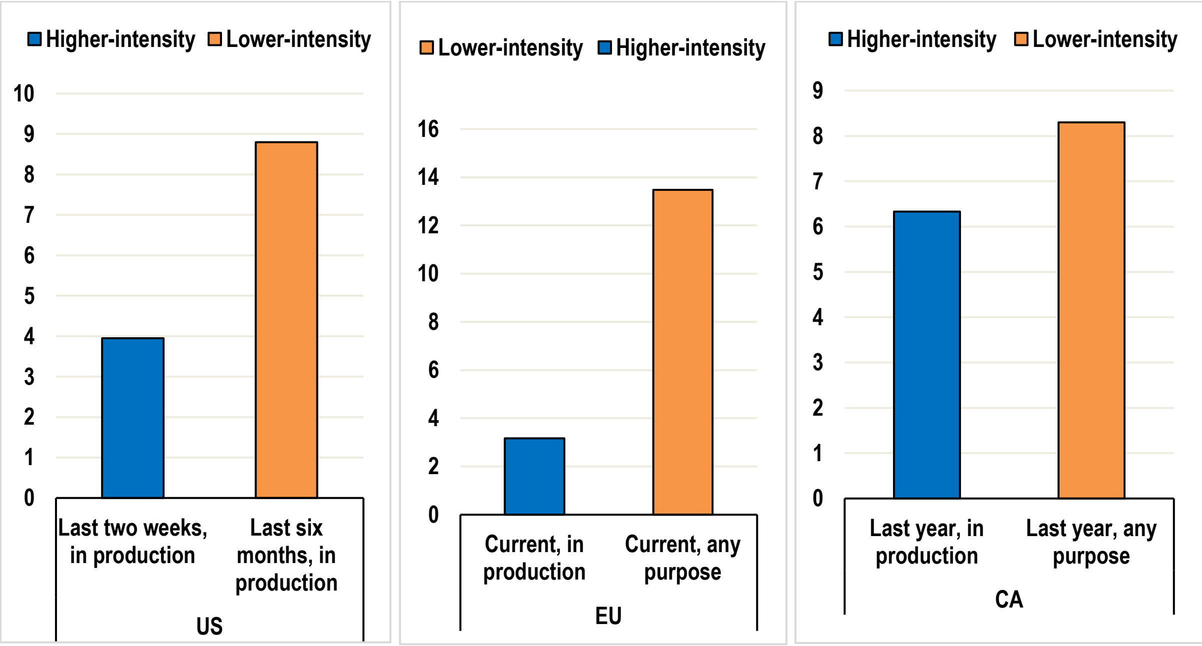
	Low intensity / ad-hoc use of AI in any business function	High intensity / regular use of AI in core business functions
Canada	Last year,	Last year,
	for any purpose	in production
	Which of the following Information and Communication Technologies (ICTs) did this business use in 20YY? […] 10: Software or hardware using artificial intelligence (AI)	Over the last 12 months, did this business or organization use Artificial Intelligence (AI) in *producing goods or delivering services*?
European Union including G7 countries France, Germany, Italy	Currently,	Currently,
	for any purpose	in production
	Does your enterprise use any of the following Artificial Intelligence (AI) technologies?	Does your enterprise use Artificial Intelligence software or systems for any of the following purposes? […] b) Use of *AI for production or service processes*
Japan	Past three years*,	Past three years*,
	for any purpose	to improve/introduce goods/services
	6-3 Usage of digitalisation (during the three years from 2019 to 2021). Please tick all boxes where they are applicable […] […] [d] Machine learning (AI) Used = improving existing goods or services; introducing new goods or services; business process automation or cost reduction; data analysis and collection, or decision-making support; others.	6-3 Usage of digitalisation (during the three years from 2019 to 2021). Please tick all boxes where they are applicable […] […] [d] Machine learning (AI) Used = *improving existing goods or services; introducing new goods or services*.
United Kingdom	Currently,	Currently,
	for any purpose	to improve operations
	Which of the following artificial intelligence technologies, if any, does your business currently use?	What does your business currently use artificial intelligence technologies for? […] -Develop a new product or service -Improve business operations
United States	Last six months,	Last two weeks,
	in production	in production
	In the last six months, what types or applications of Artificial Intelligence (AI) did this business use in producing goods or services?	Between MMM DD – MMM DD, did this business use Artificial Intelligence (AI) in *producing goods or services*?

Note: Importantly, for the United States, the question on AI adoption over the last six months stems from the AI supplement that was conducted in Q1 2024. The cited number for high intensity AI adoption in core business functions stems from the same survey wave to ensure comparability. For the rest of this document, headline figures on high intensity AI adoption in the United States refer to all survey waves in 2024.
*Referring to 2019-2021.
Source: For Canada: Statistics Canada (2024 a, b) - SDTIU (low intensity) and CSBC (high intensity); for EU: Eurostat (2025) Community survey on ICT usage and e-commerce in enterprises; for Japan: National Innovation Survey (J-NIS, 2022); for the United Kingdom: Office for National Statistics (ONS, 2025) - Business Insights and Conditions Survey (BICS); for the United States: United States Census (2025) - Business Trends and Outlook Survey (BTOS).

These surveys yield different figures of AI use across countries and AI definitions. Accordingly, numbers are substantially lower when high intensity AI adoption in core business functions is considered compared to overall adoption rates: for instance, in 2024, the respective figures are about 3% and 14% in Europe, 4 and 9% for the United States and 6 and 8% for Canada (Figure B.1). To the extent figures of the former type are available, they are used in the subsequent calculations.

Figure B.1. High-intensity adoption in core business functions is much below overall adoption

Percentage of firms reporting using AI in different time horizons and for different purposes



Notes: High intensity is defined as either very frequently or in the production of good and services are on different axis as they refer to different questions, requiring harmonisation for comparability. United States data come from United States Census BTOS, early 2025; EU data come from Eurostat, BCS, 2023 and for Canada from StatCan CSBC, 2024.

Harmonising different firm-size and sectoral coverages

Beyond differences in wording among the identified surveys, which are partially improved by focusing on high-intensity use in core business functions, there are also differences between the surveys in terms of time reference, company size coverage and sectoral coverage (Table B. 2).

Table B.2. Detailed overview of official statistical business surveys about AI use – focusing on questions related to high-intensity AI adoption in core business functions

Countries	CAN	DEU, FRA, ITA	United States
Source	StatCan, Canadian Survey on Business Conditions (CSBC)	Eurostat	United States Census, Business Trends and Outlook Survey (BTOS)
Time reference	Last 12 months	Current situation	Last two weeks
Question	Over the last 12 months, did this business or organization use Artificial Intelligence (AI) in *producing goods or delivering services*?	Does your enterprise use Artificial Intelligence software or systems for any of the following purposes? b) Use of AI for *production or service processes*	Between MMM DD – MMM DD, did this business use Artificial Intelligence (AI)1 in *producing goods or services*?
AI definition	E.g., machine learning, virtual agents, voice recognition	some of the examples may be: - predictive maintenance or process optimization based on machine learning - tools to classify products or find defects in products based on computer vision - autonomous drones for production surveillance, security or inspection tasks - assembly works performed by autonomous robots	Examples of AI: machine learning, natural language processing, virtual agents, voice recognition, etc.
Sector coverage	Excluded sectors: utilities, financial investment, management of companies/enterprises, education, several medical sub-sectors, hospitals, community food and housing, private HHs, public admin.	All activities (except agriculture, forestry and fishing, and mining and quarrying), without financial sector")	All activities except agriculture production, railroads, Postal services, monetary authorities, funds trusts and other financial vehicles, religious grant operations, private households, public admin and unclassified.
Firm-size coverage	All	10 persons employed or more	All

Source: For Canada: Statistics Canada (2024 a) - CSBC (high intensity); for EU: Eurostat (2025) Community survey on ICT usage and e-commerce in enterprises; for the United States: United States Census (2025) - Business Trends and Outlook Survey (BTOS).

Sectoral coverage

For Canada and the United States, sectors reported under the North American Industry Classification System (NAICS) were converted to the International Standard Industrial Classification (ISIC) rev. 4 (1-digit). This adjustment was made based on conversion tables that map 2-digit NAICS codes into 1-digit ISIC codes. The resulting list of considered ISIC sectors includes the entire economy except for public administration and defence; compulsory social security (O), activities of households as employers; undifferentiated goods- and services-producing activities of households for own use (T) and activities of extraterritorial organizations and bodies (U) which are also not covered by the considered surveys of Canada and the United States.[30]

As shown in Table B. 2, Eurostat does not cover all the sectors needed for an encompassing consideration of economies. For this reason, missing observations of AI adoption $AI\ adoption_{s}^{Eurostat}$ are filled by multiplying average non-missing adoption in Eurostat $AI\ adoption_{\emptyset\ Eurostat\ sectors}^{Eurostat}$ by the United States ratio

[30] A total of 18 different countries is considered: Agriculture (A), Mining (B), Manufacturing (C), Electricity (D), Water Supply (E), Construction (F), Wholesale/Retail (G), Transportation (H), Accommodation/Food (I), ICT (J), Finance (K), Real Estate (L), Prof/Science/Techn activities (M), Admin Support (N), Education (P), Health (Q), Arts Entertainment (R), and Other Services (S).

of individual sectors to the average across sectors $\frac{AI\ adoption_s^{US}}{AI\ adoption_{\emptyset\ Eurostat\ sectors}^{US}}$ that are available in both the United States and Eurostat:

$$AI\ adoption_s^{Eurostat} = AI\ adoption_{\emptyset\ Eurostat\ sectors}^{Eurostat} * \frac{AI\ adoption_s^{US}}{AI\ adoption_{\emptyset\ Eurostat\ sectors}^{US}} \quad (2)$$

Since our preferred AI adoption metric in agriculture is only available for the United States for 2023, the average ratio of ICT to agriculture in Canada and the United States has been used to interpolate the missing values for agriculture.

In the case of France, the metric for our preferred AI adoption in 2024 is neither available for the overall sectors nor for individual sectors. The only available value exists for manufacturing in 2024. Therefore, for France, missing sectoral values are derived from our preferred AI adoption measure in manufacturing, by applying the proportions that correspond to the average ratios of individual sectors relative to manufacturing in Italy and Germany.

Firm-size coverage

As shown in Table B. 2, the firm-size coverage differs across surveys. While Eurostat surveys cover firms with 10 or more employees, Canada and the United States also cover micro firms with less than 10 employees. To harmonise these differences, sectoral values for Canada and the United States are adjusted.

Since AI adoption values are not jointly available by sector and firm-size, two ratios are derived at the aggregate level: first, a multiplier capturing AI adoption by businesses with 10+ employees relative to AI adoption by businesses across all firm-size categories including micro firms, $AI\ adoption\ ratio_{total\ economy}^{C,no\ micro\ firms}$; second, the share of micro firms at the aggregate level$share\ micro\ firms_{total\ economy}^{C}$.

The next step applies these ratios at sector level. Thereby, the firm-size multiplier is applied in proportion to the respective share of micro firms in the sectors. For a given country, *c*, and sector *s*, this reads as follows (country indices compressed for simplicity):

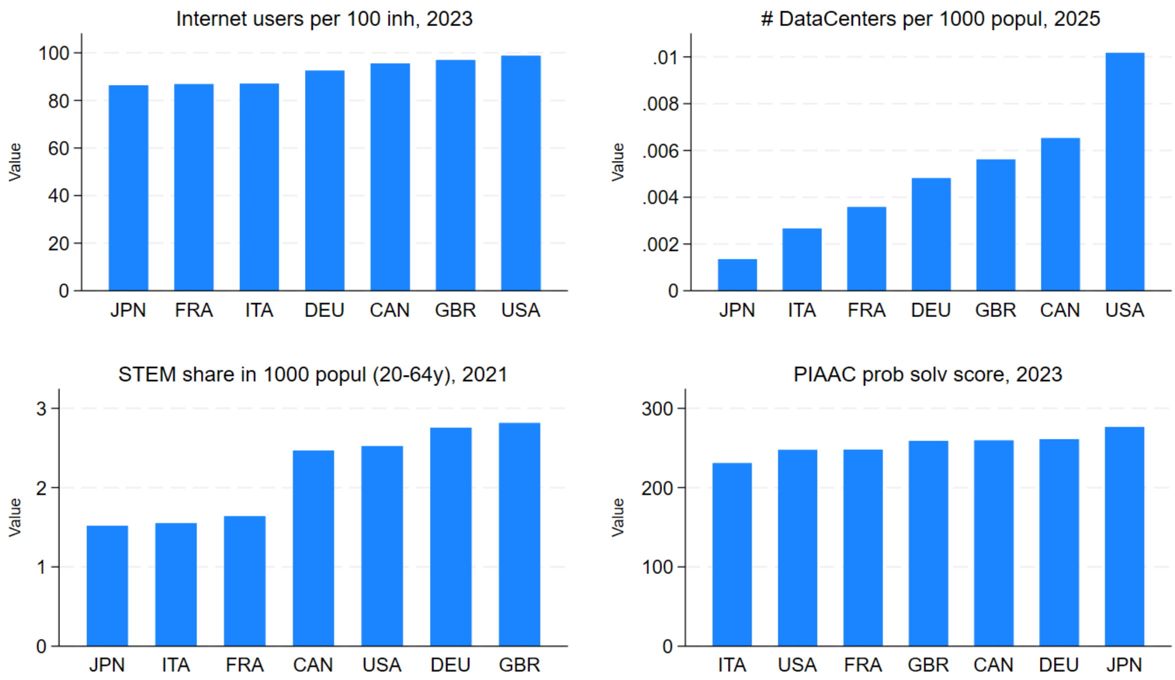
$$multiplier_{total\ economy}^{no\ micro\ firms} = \frac{AI\ adoption_{total\ economy}^{no\ micro\ firms}}{AI\ adoption_{total\ economy}^{all\ firms}} \quad (3)$$

$$AI\ adoption_s^{no\ micro\ firms} = AI\ adoption_s^{all\ firms} * \frac{share\ micro\ firms_s}{share\ micro\ fim_{total\ economy}} * multiplier_{total\ economy}^{no\ micro\ firms}$$

Generating out-of-sample cross-country prediction of AI adoption

An additional strategy to mitigate measurement related differences across countries in AI adoption is to rely on information about its underlying, structural drivers, such as digital infrastructure and skills, which are obtained from more standardized and comparable sources across countries. Relevant variables were identified using the Lasso methodology, which selects variables that are jointly most closely correlated with the dependent variable and hence are expected to be the best predictors in an out of sample context (such as the case of Japan and the United Kingdom). The exact measures for explanatory variables are also pinned down by the availability of cross-country comparable data and their ability to provide a good fit for capturing AI adoption. Digital infrastructure is measured by the share of individuals using the internet and the number of data centers per capita, while skills are represented by the proportion of STEM graduates (minimum Bachelor's degree) in the population aged 20-64, alongside PIAAC results for problem-solving abilities. Importantly, future extensions of these predictions can make use of more precise data on digital infrastructure, i.e., data centers with capacity for AI inference that is crucial for widespread Ai adoption (Russo et al., forthcoming).

Figure B.2. Descriptive statistics of considered explanatory variables



Source: ITU (2025), Data Center Map (2025), OECD (2022), OECD PIAAC (2023).

Regression results show that AI adoption is positively correlated with digital infrastructure and skills (Table B. 3). Accordingly, internet use among individuals and also the share of STEM graduates relative to the population aged 20-64 (column 1). Similarly, higher numbers of data centers and a greater capacity for problem solving among workers as measured by PIAAC are positively correlated with AI adoption across G7 sectors (columns 2 and 3).

Table B.3. The econometric link between AI adoption and its structural drivers

	(1)	(2)	(3)	(4)
VARIABLES	AI adoption	AI adoption	AI adoption	AI adoption
Internet users per 100 inhabitants, 2023	6.833***	5.659***	3.443***	3.621***
	(0.5259)	(0.8311)	(0.7652)	(0.9104)
Number of data centers per 1000 population		0.110*		-0.037
		(0.0568)		(0.0709)
STEM graduates per 1000 population, %, in 2021	0.215***	0.219***	0.478***	0.501***
	(0.0477)	(0.0475)	(0.0768)	(0.0862)
PIAAC problem solving score			3.463***	3.593***
			(0.5656)	(0.5374)
Constant	-30.589***	-24.698***	-34.441***	-36.162***
	(2.3800)	(3.9946)	(2.9420)	(4.2919)
Observations	455	455	387	387
R-squared	0.683	0.686	0.721	0.721
Sector FE	YES	YES	YES	YES
Clustering	robust	robust	robust	robust
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1				

Note: all variables are included in natural logarithms. AI adoption refers to our preferred, high-intensity measure pertaining to core business functions (see Section 2.3.1).
Source: ITU (2025), Data Center Map (2025), OECD (2022), OECD PIAAC (2023).

Based on these econometric links, out-of-sample predictions for AI adoption at the country-sector level can be generated. Due to the wider availability of data, this step is carried out based on column 2 - i.e. using the number of internet users, the number of data centres and STEM graduates as regressors. The underlying sample for these predictions comprises a total of 26 countries[31] and 18 industries[32].

This approach also helps to fill remaining data gaps for Japan and the United Kingdom where comparable, recent survey data on high intensity AI adoption in core business functions is unavailable. Moreover, to further refine available AI adoption measures, the model predictions that capture the structural capacity for AI adoption are averaged with the available AI adoption measures to obtain our preferred ultimate estimate for AI adoption among firms that captures high intensity use in core business functions (Table B.4).

[31] The following countries are considered: AUT, BEL, CAN, CZE, DEU, DNK, ESP, EST, FIN, FRA, GRC, HUN, IRL, ITA, LTU, LUX, LVA, NLD, NOR, POL, PRT, SVK, SVN, SWE, TUR, and United States.

[32] The following sectors are considered: Agriculture (A), Mining (B), Manufacturing (C), Electricity (D), Water Supply (E), Construction (F), Wholesale/Retail (G), Transportation (H), Accommodation/Food (I), ICT (J), Finance (K), Real Estate (L), Prof/Science/Techn activities (M), Admin Support (N), Education (P), Health (Q), Arts Entertainment (R), and Other Services (S).

Table B.4. AI use statistics across firms: From official headline statistics to harmonised estimates for high-intensity AI adoption in core business functions

AI use measures, in 2024*, % of businesses

Countries	AI use	*High-intensity* AI adoption in *core business functions*			
	Official headline figure	Harmonised value		Structural AI adoption capacity**	Preferred ultimate adoption estimate
		Sub-question on "high-intensity" use	*Including adjustment for firm size and sectors*	*As predicted by regressions on digital infrastructure and skills*	*Simple average of harmonised adoption and regression predictions*
CAN	6.1	6.1	7.0	4.4	**5.7**
DEU	19.8	4.9	5.7	3.9	**4.8**
FRA	9.9		4.0	2.3	**3.1**
ITA	8.2	1.9	2.4	2.1	**2.2**
JPN	4*	2*		1.9	**1.9**
United Kingdom	26.5	11***		5.4	**5.4**
United States	5.3	5.3	5.5	6.6	**6.1**

Note: *Official headline figures* refer to surveys covering any purpose of AI adoption (Eurostat: DEU, FRA, ITA; JPN, United Kingdom) or the narrower definition of AI adoption in the production of goods and services (CAN, United States). For the precise wording of respective questionnaires, see Table B.1. For the United Kingdom, official headline figures refer to values excluding firms with 0-9 employees. For Japan, the high-intensity, core business function sub-question refers to AI used to improve existing products/services or introduce new products/services. For more details, see the text in Annex B.
* Values for Japan refer to 2019-2021.
** Based on regression predictions, see Table B. 3.
*** Sub-question is not comparable with other countries, broadly focusing on "*improving operations*".
Source: For Canada: Statistics Canada (2024a) CSBC; for DEU, FRA, ITA: Eurostat (2025) Community survey on ICT usage and e-commerce in enterprises; for Japan: National Innovation Survey (J-NIS, 2022); for the United Kingdom: Office for National Statistics (ONS, 2025) - Business Insights and Conditions Survey (BICS); for the United States: United States Census (2025) - Business Trends and Outlook Survey (BTOS).

Cross validation of our preferred AI adoption estimate

To cross-check the validity of the procedure described above with alternative data sources, the rankings of AI adoption are compared with three other cross-country surveys covering AI adoption that were carried out at a similar time – the OECD Global Forum on Productivity (GFP) Employer survey on labour shortages[33], the Slack Workforce Index survey[34], and the survey on digital usage of companies by the Japanese Ministry of Internal Affairs and Communications[35]. In the context of this comparison, the main focus is on two aspects. First, the relative position of our preferred AI adoption among firms in Japan and the United Kingdom where the values rely only on the econometric regressions described above; second, the relative high position that is obtained in our preferred AI adoption measures in Germany, given its high AI values in the current paper despite concerns about its digital readiness.

[33] Respondents were asked about changes in their company related to AI use between Q2 2022 and Q2/3 2024. The survey comprised about 1,000 respondents per country, covering the G7 and other countries.

[34] The survey considers workers' AI use at work carried out in March 2024 covering 10,045 respondents across five G7 countries and Australia.

[35] The survey has been carried out during January/February 2024 in Japan, Germany, the United States and China with a total of 1,442 respondents. Respondents were asked about policies established in companies to actively utilise AI at the workplace.

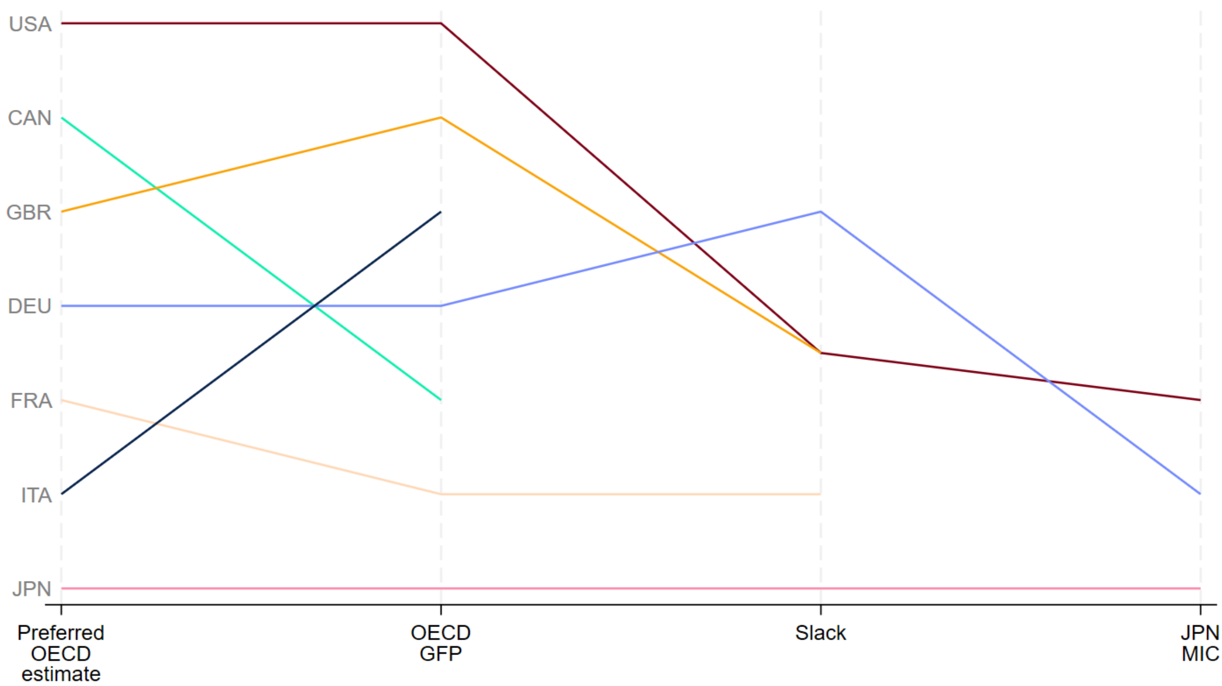
Figure A.2 shows that the United Kingdom's ranking with respect to AI adoption is close to the one of the United States across all surveys. The predicted AI adoption is ranked 3rd behind Canada and the United States which is comparable to the United Kingdom's AI adoption in the GFP survey where the United Kingdom ranks behind the United States and the Slack survey where United Kingdom's AI adoption is on a par with values from the United States.

For Japan, both the GFP and Slack survey confirm relatively low AI adoption levels in 2024. Moreover, a survey conducted by the Japanese Ministry of Internal Affairs and Communications shows that regular AI use of businesses in Japan corresponds to about one third of regular AI use of businesses in the United States – a proportion that is in line with the preferred estimates in this paper.

For Germany, the relatively high AI adoption is also confirmed by external sources. Accordingly, German AI adoption ranks similar when considering the preferred estimates and the GFP survey results. Within the Slack survey, Germany ranks first.

Figure B.3. AI adoption ranking across different surveys on AI use

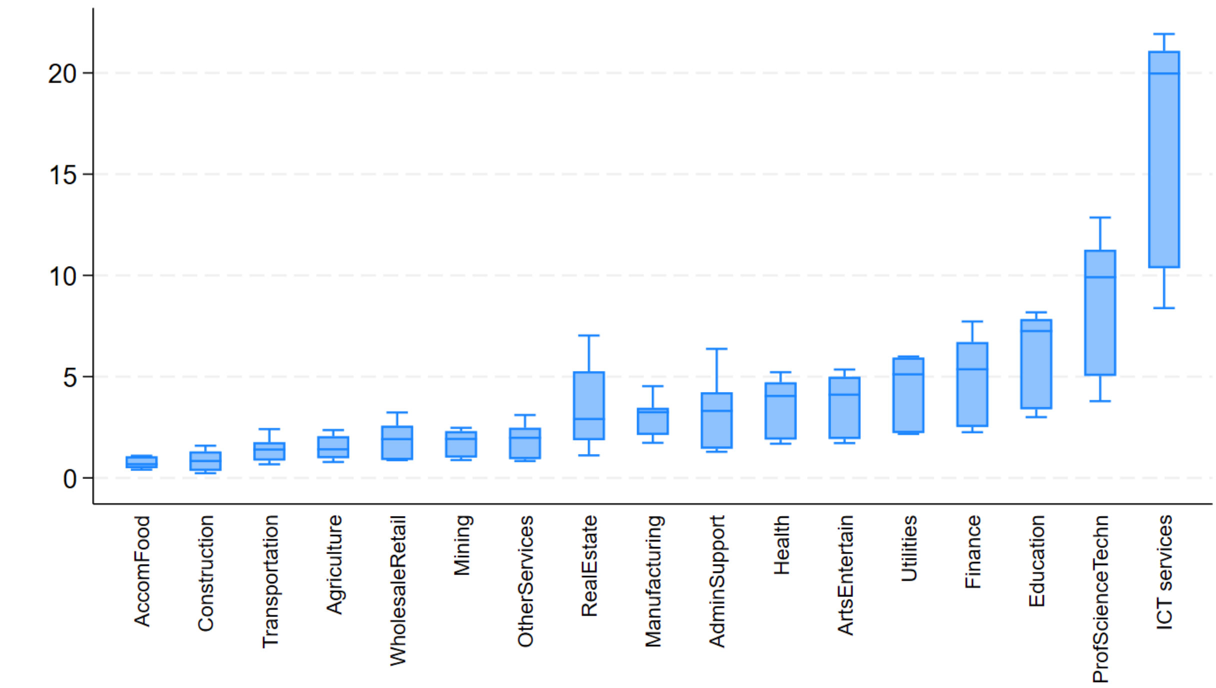
AI adoption ranking in descending order



Note: ***"OECD predicted"*** refers to AI adoption by businesses with harmonised and imputed values for Canada, Germany, France, Italy, and the United States and out-of-sample predictions for Japan and the United Kingdom. ***"OECD GFP"*** refers to AI adoption by businesses between Q2 2022 and Q2/3 2024 relying on an employer survey carried out by the OECD's Global Forum on Productivity with about 1,000 respondents per G7 country; ***"Slack"*** refers to a survey on workers' AI use at work carried out by YY in March 2024 covering 10,045 respondents across five G7 countries and Australia; ***"JPN MIC"*** refers to a survey conducted by the Japanese Ministry of Internal Affairs and Communications during January/February 2024 in Japan, Germany, the United States and China with a total of 1,442 respondents.
Source (Slack, 2024[1]; OECD, 2024a[2]; Ministry of Internal Affairs and Communications (JPN), 2024[3]).

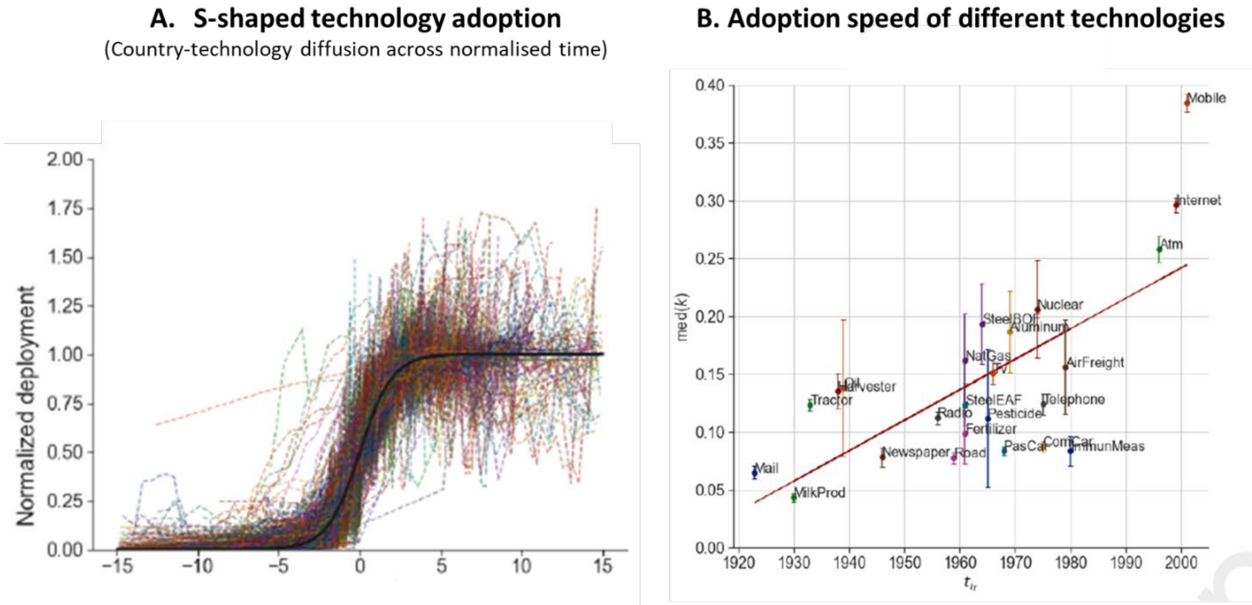
Figure B.4. Estimates of high intensity AI adoption in core business functions, by sector

In % of businesses, ranges show the distribution across countries



Source: Authors' calculations following the procedure described in Annex B.

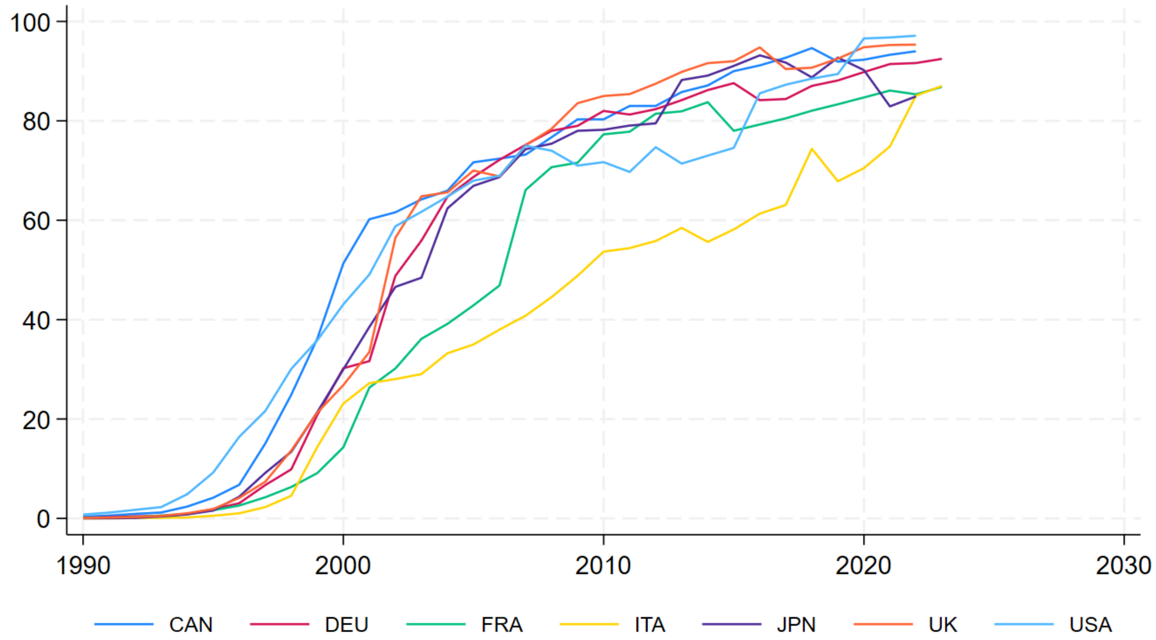
Figure B.5. Diffusion path of past technologies



Source: Tankwa et al. (2025). Panel A shows the adoption path of 3015 country-technology pairs. The data are normalized and plotted using non-dimensional coordinates to allow to comparisons across country-technology pairs with different adoption speeds and saturation levels. The thick black line traces the idealized logistic curve. Panel B shows the median adoption speed (parameter k) across countries for a given technology.

Figure B.6. S-shaped adoption path of past technologies

Individuals using the internet, in %



Source: International Telecommunication Union (2025).